Outsourcing Extinction: Economic Sophistication and the Geography of Biodiversity Loss*

Komla Avoumatsodo†   Matías Piaggio‡   Juha Siikamäki§

September 7, 2026

Abstract

Does the structure of an economy, beyond its level of income, shape biodiversity loss? We relate the Red List Index (RLI) of aggregate species extinction risk to the Economic Complexity Index (ECI) for up to 179 countries between 1993 and 2019. Conditional on income per capita, greater economic sophistication raises extinction risk within countries: a one-standard-deviation increase in the ECI lowers the RLI by one to two percent, robust to country and period fixed effects, a long-difference specification, sectoral controls, and instrumental-variables estimation using product-space diversification potential. We find little evidence that domestic land use or production intensity explains this relationship: economic sophistication does not systematically affect agricultural land use, forest cover, input intensity, or agricultural productivity. Instead, more sophisticated economies displace biodiversity pressure through trade. They are net importers of extinction risk, with a growing share of their biodiversity footprint originating from foreign production; imports from biome-sharing regions also predict subsequent declines in the domestic RLI, evidence for a range-overlap channel linking foreign habitat conversion to home-country outcomes. Sophistication, in short, does not reduce an economy's biodiversity footprint; it relocates that footprint abroad, so a cleaner domestic record reflects an illusion of territorial accounting rather than genuine abatement. Conservation policies anchored in territorial indicators therefore understate the biodiversity costs of the world's most sophisticated economies.

Keywords: Biodiversity; Extinction risk; Red List Index; Economic sophistication; Trade-embodied footprint

JEL classification: Q56; Q57; O11; O14; F18

---

*We thank colleagues at International Union for Conservation of Nature, World Bank, and University of Northern British Columbia for helpful comments. All errors are our own. The views expressed are those of the authors and do not necessarily reflect those of their institutions.

†Assistant Professor, University of Northern British Columbia: komla.avoumatsodo@unbc.ca

‡Senior Economist, World Bank Group: matias.piaggio@gmail.com

§Chief Economist, IUCN, juha.siikamaki@iucn.org

1 Introduction

Species are disappearing tens to hundreds of times faster than the average of the past ten million years, and economic activity is the proximate cause (IPBES 2019, Díaz et al. 2019, Pimm et al. 2014). The principal pressures behind biodiversity loss (land-use change, direct exploitation, climate change, pollution, and invasive species) all arise from how economies produce and consume (IPBES 2019, Maxwell et al. 2016). Yet the literature has framed this question almost entirely in terms of income: a large body of work tests for an Environmental Kuznets Curve (EKC) in biodiversity, an inverted-U in which threat first rises and then falls as countries grow richer (Naidoo & Adamowicz 2001, Asafu-Adjaye 2003, Dietz & Adger 2003, Mills & Waite 2009). This perspective focuses on the scale of economic activity. It says considerably less about its composition. Economies with similar income can differ markedly in the goods they produce and the knowledge embedded in their productive systems, and those differences may shape biodiversity pressure as much as income itself. This paper therefore asks whether the growing sophistication and technological depth of a country's productive structure shapes biodiversity threat, holding income constant.

Economic transformation changes not only how much an economy produces, but also how and what it is capable of producing. Economic complexity captures this dimension of transformation by measuring the sophistication of the productive capabilities embodied in the goods a country makes and exports (Hidalgo & Hausmann 2009, Hausmann et al. 2007, Hidalgo 2021). The Economic Complexity Index (ECI) summarizes these capabilities through the diversity and ubiquity of a country's export basket.[1] Beyond explaining differences in growth and inequality (Hausmann et al. 2007, Hartmann et al. 2017), the structure of production may also shape how economic activity bears on natural ecosystems. As economies become more sophisticated, production typically shifts away from a narrow base of resource- and land-intensive activities toward a broader portfolio of knowledge-intensive industries. This shift could, in principle, ease pressure on nature, consistent with evidence that greater sophistication lowers carbon emissions and supports cleaner production in advanced economies (Romero & Gramkow 2021, Mealy & Teytelboym 2022, Stojkoski et al. 2023). We find instead that greater sophistication raises biodiversity threat, suggesting that how and what economies produce matter as much as how much they produce.

Relating the Red List Index (RLI) of aggregate extinction risk (Butchart et al. 2004, 2007) to the ECI for up to 179 countries between 1993 and 2019, we first find that greater economic sophistication increases extinction risk among a country's own species. The estimate is robust to country and period fixed effects, to a long-difference design that removes all time-invariant national heterogeneity, to controlling for the full sectoral composition of output, and to the temporal aggregation of the data. Reverse causation is unlikely, because export structure is predetermined: it is built from capabilities accumulated over decades, not from the current status of a country's species.[2] Instrumenting sophistication with its

[1] This capability-based measure differs from the sectoral reallocation—from agriculture toward manufacturing and services—emphasized in the classical literature on structural change.

[2] Economic complexity reflects a country's stock of productive capabilities accumulated over decades. In contrast, changes in the conservation status of species have only indirect pathways to influence productive structures. For biodiversity loss to generate the observed relationship, it would need to affect industrial capabilities, innovation, skills, and export composition sufficiently to change ECI. This is a much slower and less direct process than the effect of structural transformation on ecosystems.

product-space diversification potential leaves a negative effect of similar or larger magnitude on a strong first stage. Sophistication, on this evidence, does not appear to spare a country's own biodiversity.

Our second result is that this effect does not run through the obvious domestic channel. If more sophisticated economies threatened their own species, one would expect the effect to operate through their own land use: more cropland, less forest, more intensive fertilizer and pesticide use, higher agricultural productivity, or faster deforestation. None of these domestic channels accounts for the sophistication effect. Across every specification the sophistication coefficient is unchanged, the channels are not themselves moved by sophistication, and none mediates the relationship. The pressure that sophistication exerts on biodiversity is not visible in a country's own land. This does not make land use irrelevant; it suggests that the land being converted lies abroad.

Our third result locates that pressure abroad. A growing literature traces the species threat and extinction risk embodied in international supply chains, showing that consumption in one country drives biodiversity loss in others (Lenzen et al. 2012, Moran & Kanemoto 2017, Marques et al. 2019, Wilting et al. 2017, Chaudhary & Kastner 2016). Using a bilateral matrix of extinction-risk footprints (Irwin et al. 2022), we find that more sophisticated economies are net importers of extinction risk: they source a growing share of their biodiversity footprint from other countries' species. Economies in the top tercile of sophistication place about seventy percent of their consumption footprint abroad and are net importers of extinction risk, while the least sophisticated economies place far less abroad and are net exporters. Sophistication, on this reading, does not reduce the biodiversity cost of consumption so much as relocate it to the ecosystems of trading partners. Familiar commodities make the point concrete: oil palm, soy, and the metals of the energy transition each convert biodiverse habitat far from where the final good is consumed, yet none of these costs appears in the importing country's domestic land accounts.

Our fourth result closes the loop back to the country's own biodiversity status. Net-importer status by itself is not sufficient: sophistication raises a country's own extinction risk only if the foreign habitat converted on its behalf also feeds back through species ranges it shares with that habitat, rather than simply displacing risk onto unrelated species elsewhere. We test this feedback channel directly, using a species-range overlap matrix built from IUCN Red List and BirdLife International range polygons for the four taxa, mammals, birds, amphibians, and reef-building corals, that constitute the Red List Index itself. Countries that import a larger share of their extinction-risk footprint from range-overlapping trading partners subsequently experience a significantly larger decline in their own Red List Index, a relationship that, if anything, strengthens once the domestic land-use channel is controlled for directly. A measure built from actual species occurrence also outperforms a coarser same-continent classification, indicating that the channel is genuinely ecological rather than a proxy for geographic or cultural proximity between trading partners.

Taken together, these four results tell a single story: sophistication can make an economy look cleaner at home, but that cleanliness is an illusion of territorial accounting, and the extinction risk it appears to erase is instead relocated abroad. This reframes a debate that has centered on income. The biodiversity-EKC literature, by relying on cross-sectional income variation, compares the experience of poor and rich countries, in which the relocation of damage across borders can be hard to distinguish from genuine abatement (Stern 2004, Vincent 1997).[3] The complexity-and-environment literature has

[3]The same logic applies to economic sophistication: more sophisticated economies appear, across countries, to face less species threat, even as becoming more sophisticated coincides with rising threat within a country over time.

established that sophistication can lower territorial emissions (Romero & Gramkow 2021) even as it raises aggregate ecological pressure (Neagu 2020), and the corresponding question for biodiversity in particular remains open. We find that sophistication does not deliver biodiversity decoupling, because biodiversity loss travels through trade in a way that carbon accounting within borders does not capture (Lenzen et al. 2012, Wiebe & Wilcove 2025, Li & Chen 2024). The policy implication follows directly: conservation targets and disclosure frameworks anchored in domestic indicators will tend to understate the biodiversity cost of the world's most sophisticated economies.

The framework that ties these results together brings the trade-and-environment model of Copeland & Taylor (1994, 2004) into contact with the species–area relationship from ecology (Preston 1962, Pimm & Raven 2000, Brooks et al. 2002) and economic complexity as a source of comparative advantage and productivity (Hidalgo & Hausmann 2009, Hausmann et al. 2007). The key link is the species-range overlap matrix, ω_{ij}, which measures the extent to which the species occurring in country i also depend on habitat in country j.[4] Through this ecological bridge, habitat conversion embodied in trade propagates extinction risk across borders whenever species ranges span multiple countries, generating the biodiversity measure built directly from the conservation status of the species present in each country. None of the results above emerges from trade theory, species–area relationships, or economic complexity in isolation; each requires the three integrated into a single framework.

Related Literature. The framework naturally connects three literatures that have evolved largely in parallel: the empirical literature on economic growth and biodiversity loss, the economics of productive complexity, and the trade-embodied accounting of biodiversity footprints. This paper contributes to each. The first asks whether growth threatens or spares biodiversity, and frames the question in terms of the level of income. Borrowing the Environmental Kuznets Curve logic developed for pollution, a large empirical literature looks for an inverted-U between income and species threat, with threat rising among poor countries and falling once they are rich enough (Naidoo & Adamowicz 2001, Asafu-Adjaye 2003, Dietz & Adger 2003, Mills & Waite 2009). The evidence is mixed, and it rests largely on cross-country comparisons of poor and rich nations at a point in time. Stern (2004) noted for carbon that such cross-sectional curves can in part reflect confounding or the relocation of damage rather than its abatement, a caution that may apply with particular force to biodiversity, whose damage is fixed to the converted habitat. Our approach differs from this strand in both what we ask and how we identify it. We hold income constant and vary the structure of production, and we estimate the relationship within countries over time, where sophistication raises extinction risk even though the cross-section points the other way.

The second strand is the economics of complexity. Building on the product space and the idea that what a country exports reveals the capabilities it has accumulated, this literature measures economic complexity from the diversity and ubiquity of export baskets (Hidalgo & Hausmann 2009, Hausmann et al. 2007, Hidalgo 2021) and shows that it predicts future growth and inequality beyond what factor accumulation explains (Hartmann et al. 2017, Mealy & Teytelboym 2022). A more recent branch turns to the environment and finds, for carbon, that more complex economies emit less per unit of output and innovate toward cleaner technologies, so that sophistication and territorial decarbonization tend to move together (Romero & Gramkow 2021, Boleti et al. 2021). The exception is Neagu (2020), who documents that the aggregate ecological footprint rises with complexity, hinting that the carbon-

[4] We construct Ω empirically from species-level range data. Appendix D.1 describes the construction in full.

decoupling pattern may not generalize to land- and habitat-based environmental indicators. We carry this literature to biodiversity and find that the environmental dividend does not generalize: sophistication raises extinction risk. The contrast with carbon need not be a puzzle, because the two outcomes are recorded differently once production shifts abroad. A country's territorial carbon emissions fall when land- and emission-intensive production moves abroad, with no mechanism to record that shift back into the country's own emissions total. Biodiversity does not work the same way: because species ranges cross borders, habitat conversion abroad can still register in a country's own Red List Index whenever the affected species also occur at home. Shifting production abroad lowers a country's territorial carbon footprint outright; it does not, in general, lower its own biodiversity level by the same logic.

The third strand quantifies the biodiversity loss embodied in international trade, an inquiry whose theoretical roots lie in the trade-and-environment framework of Copeland & Taylor (1994, 2004). Using multi-regional input–output and bilateral footprint accounts, this literature shows that consumption in wealthy economies drives a large and rising share of the species threat recorded in their trading partners, concentrated in the biodiverse tropics,[5] with commodity-specific evidence on the biodiversity costs of trade reinforcing the macro pattern (Dalheimer et al. 2024) and recent environmental-economics work documenting trade-driven leakage at the country level (Holladay et al. 2018, Abman et al. 2026, Carattini et al. 2026). A complementary line of work uses sub-national variation within countries to relate local economic activity to species imperilment (Liang et al. 2021). Both literatures establish the geography of the problem and are largely descriptive in focus, mapping where consumption-driven biodiversity loss falls. We build on them by adding two further questions: which structural feature of an economy makes it a large net exporter of biodiversity pressure, and whether the footprint a country imports is reflected in the extinction risk it records at home. We show that economic sophistication is a systematic structural driver of how much biodiversity pressure a country displaces abroad, and that the most externalized economies tend to be those whose own species grow more threatened. A trade model in which sophistication shifts comparative advantage toward land-saving goods, joined to a species–area map with cross-border ranges, draws these facts together and frames the policy question of consumption-based accountability (Vergez et al. 2025, Dasgupta 2021).

Outline. The remainder of the paper is organized as follows. Section 2 develops the theoretical framework and derives its testable predictions. Section 3 describes the data and empirical strategy. Section 4 presents the empirical evidence, examining the relationship between economic sophistication and biodiversity threat, the role of trade-embodied extinction risk, and a range of robustness and mechanism analyses. Section 5 discusses the implications of the findings for the biodiversity-growth literature and conservation policy. Section 6 concludes.

2 Theoretical framework

This section develops a theoretical framework that links trade and economic sophistication to biodiversity loss and provides the foundation for the paper's empirical analysis. We build on the trade-and-environment framework of Copeland & Taylor (1994, 2004) and augment it with a species–area relation-

[5] See, for example, Lenzen et al. (2012), Moran & Kanemoto (2017), Marques et al. (2019), Wilting et al. (2017), Chaudhary & Kastner (2016), Irwin et al. (2022), Wiebe & Wilcove (2025).

ship linking habitat conversion to extinction risk (Pimm & Raven 2000, Brooks et al. 2002). The model yields comparative-statics predictions that map directly onto the empirical specifications and clarifies how greater sophistication can affect biodiversity through changes in the composition and location of production. We then characterize the inefficiency of the decentralized equilibrium and derive the policy instrument that restores the efficient allocation.

2.1 Environment

The economy is composed of N countries, indexed by $i = 1,\dots,N$, each with population $n_i > 0$ that supplies a fixed labor endowment L_i, inelastically supplied and mobile across sectors within the country but not across countries. Countries trade two final goods x and y. Good x is habitat-intensive: producing it converts natural land that would otherwise shelter species (agriculture, pasture, and resource extraction). Good y is knowledge-intensive and land-saving (sophisticated manufactures and services). Each country i is endowed with a stock of convertible habitat $\bar{A}_i$. For the comparative-statics results below, it is convenient to fix one country as Home, denoted i, and to write $-i$ for the rest of the world, the aggregate of every other country $j \neq i$.

Preferences. Preferences are identical and homothetic. A representative consumer in country i values the two goods and the biodiversity present in its territory,

$$U_i = \lambda \log c_i^x + (1-\lambda)\log c_i^y + \phi\, b_i, \qquad \lambda \in (0,1),\ \phi \geq 0, \tag{2.1}$$

where c_i^x, c_i^y are consumption levels,[6] b_i is the conservation status of the species present in country i (defined in Section 2.2), and ϕ is the weight placed on it. Cobb–Douglas preferences over goods fix expenditure shares, so that $p_x c_i^x = \lambda V_i$ and $p_y c_i^y = (1-\lambda)V_i$, where V_i is income and (p_x, p_y) are world prices for goods x and y. This is the property that lets the model speak to regressions that condition on income: at a given income and given prices, consumption is pinned down, so sophistication can act only through the composition and the location of production.

Technology. Technology uses labor under diminishing returns, with each sector employing a fixed complementary factor (knowledge capital in y, land in x), following a specific-factors structure:

$$y_i = \kappa_i\,(L_i^y)^\eta, \qquad x_i = B\,(L_i^x)^\eta, \qquad L_i^x + L_i^y = L_i, \qquad \eta \in (0,1). \tag{2.2}$$

B is the productivity of the habitat-intensive sector, common across countries, and η is the labor elasticity shared by both sectors, governing diminishing returns to the sector-specific factor. The parameter κ_i is economic sophistication: the diversity and technological depth of a country's capabilities, which raise productivity in the knowledge-intensive sector. This is the reduced-form image of the product space of Hidalgo et al. (2007), in which more sophisticated economies command the capabilities to produce

[6] We read c_i^x as the habitat-intensive content embodied in country i's final consumption, not literal purchases of an unprocessed commodity. Consumption-based footprint accounting, the convention behind the empirical measure used in Section 4.3 (Irwin et al. 2022, Lenzen et al. 2012), traces a commodity through however many production stages it passes and assigns the accumulated impact to whoever consumes the finished good, so c_i^x already nets out any use of the habitat-intensive good as an input further upstream.

many advanced, land-saving goods. Higher κ_i raises the relative return to allocating labor to y and therefore the relative cost of producing x, giving more sophisticated economies a comparative advantage in land-saving goods.

Producing the habitat-intensive good converts land in fixed proportion, $h_i = \gamma_i x_i$, where h_i is hectares of habitat lost and $\gamma_i > 0$ its land intensity. We interpret h_i broadly. Cropland cleared for export agriculture is the most direct case, but the same logic covers the forest opened by logging, the ground broken by mining for the metals of the green transition, the rivers and soils contaminated by industrial and agricultural runoff, and the roads and ports laid through ecosystems to move what these activities produce.

2.2 Habitat, species, and conservation status

Converted habitat drives extinction through the species–area relationship, one of the most robust regularities in ecology (Pimm & Raven 2000, Brooks et al. 2002). With original habitat $\bar{A}_i$ and conversion h_i, the surviving habitat is $A_i = \bar{A}_i - h_i$, and the share of species in country i committed to extinction is

$$E_i = 1 - \left(\frac{A_i}{\bar{A}_i}\right)^z = 1 - \left(1 - \frac{h_i}{\bar{A}_i}\right)^z, \qquad z \in (0,1), \tag{2.3}$$

increasing and convex in conversion, with the canonical exponent $z \approx 1/4$ derived theoretically by Preston (1962) and routinely used in habitat-loss extinction predictions (Brooks et al. 2002, Pimm & Raven 2000).

Species ranges cross borders, so the biodiversity status of species present in one country depends on habitat loss in others. Let $\Omega = (\omega_{ij})_{1\leq i,j\leq N}$ denote the species-range overlap matrix, where each element $\omega_{ij} \geq 0$ is the share of country i's species assemblage whose ranges include country j. Rows sum to one, $\sum_j \omega_{ij} = 1$. The term ω_{ii} captures locally endemic species, while ω_{ij} for $j \neq i$ captures cross-border overlap arising from shared, migratory, or biome-spanning ranges. The biodiversity level of country i is the overlap-weighted share of its species not yet committed to extinction,

$$b_i = 1 - \sum_{j=1}^{N} \omega_{ij} E_j = \sum_{j=1}^{N} \omega_{ij} \left(1 - \frac{h_j}{\bar{A}_j}\right)^z. \tag{2.4}$$

The marginal effect of conversion in country j on the biodiversity status of country i is

$$\frac{\partial b_i}{\partial h_j} = -\,\omega_{ij} m_j, \qquad m_j \equiv \frac{z}{\bar{A}_j}\left(1 - \frac{h_j}{\bar{A}_j}\right)^{z-1} > 0, \tag{2.5}$$

where m_j is the marginal extinction intensity of country j: the species lost per hectare converted. It is larger where habitat is biodiverse and intact, so conversion relocated into species-rich frontiers is especially damaging. The empirical footprint accounts show that the consumption of rich economies falls disproportionately on biodiverse tropical habitat (Lenzen et al. 2012, Chaudhary & Kastner 2016, Irwin et al. 2022).

Equation (2.4) expresses b_i as a weighted sum of countries' survival shares, $g_k \equiv (1 - h_k/\bar{A}_k)^z$. Because the same g_k can enter the biodiversity levels of multiple countries whenever their species ranges overlap, the biodiversity levels of those countries are mechanically linked. The resulting dependence is therefore a consequence of the spatial structure of species ranges, rather than an incidental correlation across countries.

Proposition I (Spatial dependence)**.** *The cross-country biodiversity levels* $\{b_i\}_{i=1}^N$ *are positively spatially correlated whenever range overlap is non-trivial. Treating* g_k *as random across countries with finite variance,*[7]

$$\mathrm{Cov}(b_i, b_j) \;=\; \sum_{k,l} \omega_{ik}\, \omega_{jl}\, \mathrm{Cov}(g_k, g_l) \;\geq\; 0, \tag{2.6}$$

with strict inequality whenever there exists k such that $\omega_{ik}\,\omega_{jk} > 0$.

Shared ranges make a country's biodiversity status partly a function of its neighbors'. When habitat is converted in one country it commits species to extinction wherever their range extends, so countries that share biogeographic space share biodiversity outcomes. Proposition I states this formally: biodiversity levels are positively correlated with the levels of the countries they overlap with.

2.3 Trade equilibrium and the composition of production

Take world prices as given and normalize $p_y = 1$, writing $p \equiv p_x$ for the relative price of the habitat-intensive good. Cost minimization equates the value marginal product of labor across sectors,

$$\kappa_i\, \eta\, (L_i^y)^{\eta-1} \;=\; p\, B\, \eta\, (L_i^x)^{\eta-1}. \tag{2.7}$$

Cancelling η and collecting the labor ratio on one side,

$$\left(\frac{L_i^y}{L_i^x}\right)^{\eta-1} \;=\; \frac{pB}{\kappa_i}. \tag{2.8}$$

Since $\eta \in (0,1)$, the exponent $\eta - 1$ is negative; raising both sides to the power $1/(\eta-1)$ inverts the ratio on the right-hand side and gives

$$\frac{L_i^y}{L_i^x} \;=\; \left(\frac{\kappa_i}{pB}\right)^{\frac{1}{1-\eta}}. \tag{2.9}$$

Substituting $L_i^y = (L_i^y/L_i^x)\, L_i^x$ into the labor constraint $L_i^x + L_i^y = L_i$ and dividing through by L_i delivers the sectoral labor shares,

$$\frac{L_i^x}{L_i} \;=\; \frac{1}{1 + (\kappa_i/pB)^{1/(1-\eta)}}, \qquad \frac{L_i^y}{L_i} \;=\; \frac{(\kappa_i/pB)^{1/(1-\eta)}}{1 + (\kappa_i/pB)^{1/(1-\eta)}}. \tag{2.10}$$

Definition 1 (Economic transformation)**.** *Country i's economic transformation is the reallocation of its labor share across sectors induced by rising sophistication:* $d(L_i^y/L_i)/d\kappa_i > 0$ *and* $d(L_i^x/L_i)/d\kappa_i < 0$.

By Equation (2.10), the habitat-intensive share L_i^x/L_i is strictly decreasing in κ_i while the knowledge-intensive share L_i^y/L_i rises by exactly the same amount: a shift of productive structure toward knowledge-intensive activity and away from habitat-intensive activity.

[7] g_k is random because sophistication κ_k is treated as an exogenous cross-country primitive. By Lemma 1, g_k is a deterministic function of κ_k and therefore inherits its cross-country distribution. See Appendix A.4 for the proof that $\mathrm{Cov}(g_k, g_l) \geq 0$ across countries.

Lemma 1 (Composition effect)**.** *At given prices, the supply of the habitat-intensive good and the habitat converted to produce it are strictly decreasing in sophistication:* $\partial x_i/\partial\kappa_i < 0$ *and* $\partial h_i/\partial\kappa_i = \gamma_i\,\partial x_i/\partial\kappa_i < 0$.

The proof is immediate from Equation (2.10): L_i^x/L_i falls in κ_i, and since L_i is fixed, so does the level $L_i^x = L_i\cdot(L_i^x/L_i)$; then $x_i = B(L_i^x)^\eta$ and $h_i = \gamma_i x_i$ fall as well. The explicit derivative is in Appendix A.1.

Countries trade to clear world markets. Consider a rise in the sophistication of country i, holding every other country's κ_j fixed. By Lemma 1, country i responds by producing relatively more y and less x. World-market clearing for the habitat-intensive good,

$$\sum_{j=1}^{N} x_j(p,\kappa_j) \;=\; \sum_{j=1}^{N} c_j^x, \tag{2.11}$$

together with balanced trade, determines the equilibrium price p. The contraction in country i's supply of x raises its relative supply of y and, in equilibrium, raises the relative price p of the habitat-intensive good. This price increase transmits the effect of country i's greater sophistication to its trading partners. Although their κ_j's are unchanged, each country's supply of x is increasing in p, by the same cost-minimization logic as in Lemma 1 applied to p rather than κ. Hence, every trading partner expands its production of x, rather than the response being confined to the rest of the world in aggregate. Country i, in turn, meets a larger share of its fixed consumption c_i^x through imports. Habitat conversion therefore shifts from country i toward each of its trading partners as i becomes more sophisticated: $dh_i/d\kappa_i < 0$ and $dh_j/d\kappa_i > 0$ for every $j \neq i$ individually (see Appendix A.2 for details).

2.4 Sophistication and extinction risk

Define the conditional effect of sophistication as the change in country i's biodiversity status holding income V_i and prices fixed,

$$\left.\frac{d\,b_i}{d\kappa_i}\right|_{V_i} = -\,\omega_{ii}\,m_i\,\frac{dh_i}{d\kappa_i} \;-\; \sum_{j\neq i}\omega_{ij}\,m_j\,\frac{dh_j}{d\kappa_i}, \tag{2.12}$$

obtained by differentiating Equation (2.4) and using Equation (2.5) (see Appendix A.3). The first term captures the domestic channel, and the second, summed over every country whose species ranges overlap with country i's, the foreign channel. Holding V_i and prices fixed isolates the effect of sophistication operating through the composition and location of production.

The two channels work in opposite directions, and not for symmetric reasons. By Lemma 1 and Appendix A.2, greater sophistication reduces country i's own production of the habitat-intensive good net of the price feedback it induces, so $dh_i/d\kappa_i \leq 0$. The domestic term is therefore non-negative: at higher sophistication, country i converts less habitat than it would under the lower-sophistication counterfactual. Because habitat conversion is irreversible in the model, this effect represents less further conversion rather than recovery of habitat already lost.

The foreign channel points the other way, and describes a real, additional loss rather than an avoided one. World demand for the habitat-intensive good has not fallen, so as country i's domestic production contracts, the shortfall is met through imports, inducing its trading partners to expand production and habitat conversion of their own, $dh_j/d\kappa_i > 0$ for every $j \neq i$. Where species ranges overlap, that additional conversion becomes extinction risk country i's own status counts too. The total in Equation (2.12) is therefore ambiguous: sophistication leaves country i with less habitat converted at home than the counterfactual, but can still raise its own extinction risk through additional conversion abroad.

Proposition II (The foreign channel of extinction risk)**.** *Consider country i once it has fully specialized away from the habitat-intensive good: it no longer converts any land in that sector, yet keeps sophisticating further. Then*

$$\left.\frac{d\,b_i}{d\kappa_i}\right|_{V_i,\,h_i=0} = -\sum_{j\neq i}\omega_{ij}\,m_j\,\frac{dh_j}{d\kappa_i} < 0, \tag{2.13}$$

strictly negative because $dh_j/d\kappa_i > 0$ for every $j \neq i$.

The mechanism is straightforward. As sophistication rises, country i's trading partners keep converting more of their own land to supply what country i no longer produces. Wherever its species overlap with those in its trading partners, this additional conversion contributes to the extinction risk reflected in its biodiversity status. Once domestic conversion has stopped, further sophistication can only worsen its biodiversity level: with the domestic term at zero, nothing remains to offset the foreign channel.[8]

The key to this mechanism is that habitat conversion is determined by production, not consumption. Country i's conversion is $h_i = \gamma_i x_i$; its consumption of the habitat-intensive good is $c_i^x = \lambda V_i/p$. Holding income and the world price fixed, c_i^x does not vary with κ_i: sophistication does not change country i's demand for the good, only where that demand is supplied. By Lemma 1, higher sophistication reduces domestic production, widening the gap between consumption and production, and that gap is filled by imports.

That reallocation has a direct implication for the geography of extinction risk embodied in trade: as domestic production is replaced by imports, more of the extinction risk in country i's consumption should originate abroad, and its trade position should tilt from net exporter to net importer of extinction risk as it sophisticates. The next proposition states this formally.

Proposition III (Leakage and net imports of extinction risk)**.** *Let μ denote the extinction intensity of a unit of the habitat-intensive good produced abroad, analogous to m_i for production in country i. Define country i's net imported extinction risk as the extinction risk generated abroad to satisfy its domestic consumption, net of the extinction risk generated at home to produce its exports,*

$$v_i = \underbrace{\mu\max(c_i^x - x_i,\,0)}_{\textit{risk generated abroad}} - \underbrace{\gamma_i m_i \max(x_i - c_i^x,\,0)}_{\textit{risk embodied in exports}}. \tag{2.14}$$

The portion of domestic production consumed at home, $\min(x_i, c_i^x)$, *generates* $\gamma_i m_i \min(x_i, c_i^x)$ *in domestic extinction risk. Thus, the total extinction risk embodied in country i's consumption is* $\gamma_i m_i \min(x_i, c_i^x) + \mu\max(c_i^x - x_i, 0)$*, and the share of that risk generated abroad is*

$$s_i = \frac{\mu\max(c_i^x - x_i,\,0)}{\gamma_i m_i \min(x_i,\,c_i^x) + \mu\max(c_i^x - x_i,\,0)}. \tag{2.15}$$

[8]Sophisticated production can also require more land-intensive activity, critical-mineral extraction, siting for knowledge-intensive industries, or non-traded habitat-intensive goods responding to domestic income rather than trade, mechanisms Lemma 1 does not capture. Proposition II is therefore a prediction of one specific production-composition mechanism.

Then $ds_i/d\kappa_i > 0$ *and* $d\nu_i/d\kappa_i > 0$.

The two terms in Equation (2.14) are, respectively, the extinction risk generated abroad when domestic consumption exceeds domestic production, and the extinction risk generated domestically to produce the portion of output that is exported. Both results follow from the same two facts established above: c_i^x does not vary with κ_i, while x_i strictly decreases. Once κ_i is sufficiently high that $x_i < c_i^x$, country i becomes an importer and supplies the gap $c_i^x - x_i$ through foreign production. In this regime, s_i is the share of the fixed consumption basket whose associated extinction risk is generated abroad. As x_i falls, the imported quantity rises while domestic production falls, so s_i increases unambiguously. At the same time, $\nu_i = \mu(c_i^x - x_i)$ rises as x_i falls. For a less-sophisticated country that remains an exporter, $x_i > c_i^x$, so $\nu_i = -\gamma_i m_i (x_i - c_i^x)$; as domestic production falls, the extinction risk embodied in its exports declines, causing ν_i to rise toward zero. Thus, $d\nu_i/d\kappa_i > 0$ holds on both sides of the export–import threshold, while s_i rises once the country enters the importing regime. As sophistication increases, country i therefore moves from exporting extinction risk to importing it.

2.5 Welfare and policy

The cross-border nature of the biodiversity externality makes the decentralized allocation inefficient. A unit of habitat conversion in country j affects the biodiversity of every country whose species ranges overlap with j, so its social cost depends on the full set of weights $\{\omega_{ij}\}_{i=1}^N$. To characterize the resulting wedge, consider a planner who chooses the world allocation $\{c_i^x, c_i^y, L_i^x, L_i^y\}_{i=1}^N$ to maximize world welfare

$$\mathscr{W} = \sum_i n_i \left[\lambda \log c_i^x + (1-\lambda)\log c_i^y + \phi\, b_i\right], \tag{2.16}$$

subject to the production technology in Equation (2.2), the habitat-conversion identity $h_i = \gamma_i x_i$, the biodiversity measure in Equation (2.4), and goods-market clearing,

$$\sum_i c_i^x = \sum_i x_i \qquad \text{and} \qquad \sum_i c_i^y = \sum_i y_i.$$

The planner's first-order condition with respect to h_j gives the social marginal cost of conversion in country j,

$$\mathrm{SMC}_j \equiv -\sum_i n_i \phi \frac{\partial b_i}{\partial h_j} = \phi\, m_j \sum_i n_i \omega_{ij}. \tag{2.17}$$

The summation captures the fact that a unit of conversion in j can reduce biodiversity in several countries through overlapping species ranges.

In the decentralized allocation, this full cost is not internalized. If consumption in country k induces conversion in country j, the private marginal cost reflects only the biodiversity loss borne in k,

$$\mathrm{PMC}_{kj} = \phi\, n_k \omega_{kj} m_j. \tag{2.18}$$

The resulting wedge is

$$\Delta_{kj} \;\equiv\; \mathrm{SMC}_j - \mathrm{PMC}_{kj} \;=\; \phi\, m_j \sum_{i \neq k} n_i\, \omega_{ij} \;\geq\; 0. \tag{2.19}$$

It measures the biodiversity damage borne by countries other than the country whose consumption induces the conversion.

Trade determines where this unpriced damage is realized. By Proposition III, greater sophistication reduces habitat-intensive production at home and shifts part of the associated demand toward foreign producers. Conversion in country j can then affect countries beyond the importing country k whenever species ranges extend across their borders. The externality therefore follows the geography of species ranges rather than the geography of the transaction. Corrective policy must consequently account for the full damage generated by conversion at its source.

Proposition IV (Efficient taxation)**.** *The decentralized equilibrium is inefficient whenever $\Delta_{kj} > 0$ for some country pair (k, j). A territorial Pigouvian tax,*

$$t_j \;=\; \phi\, n_j\, \omega_{jj}\, m_j, \tag{2.20}$$

prices only the biodiversity damage borne within country j and leaves the cross-border component $\phi\, m_j \sum_{i \neq j} n_i\, \omega_{ij}$ unpriced. The first-best allocation is instead implemented by

$$\tau_j \;=\; \phi \sum_i n_i\, \omega_{ij}\, m_j, \tag{2.21}$$

levied on each unit of habitat conversion in country j, regardless of where the resulting output is consumed. Equivalently, the same allocation can be decentralized through transfers from consuming countries to country j. Let x_j^k denote the quantity of j's output consumed by country k, so that $x_j = \sum_k x_j^k$, and let $\theta_{kj} \equiv x_j^k / x_j$ denote its share. A transfer

$$T_{kj} \;\equiv\; \theta_{kj}\, h_j\, \Delta_{kj} \tag{2.22}$$

from country k to country j for the conversion associated with k's consumption also implements the first-best.[9]

The transfer in Equation (2.22) charges each consuming country for the part of the uninternalized damage attributable to its own demand. Summed over consuming countries, these transfers cover the gap between the full social cost of conversion and the biodiversity cost already internalized privately.

The distinction between t_j and τ_j is therefore one of scope. The former prices the damage that conversion in j imposes on j itself; the latter prices the damage borne across all countries connected to j through species-range overlap. Because τ_j enters the cost of producing x_j, it affects the price of that output wherever it is sold. The extent to which this cost is passed through to consumers, rather than absorbed by producers in j, depends on relative supply and demand elasticities, which the model does not determine. Under τ_j, however, the marginal cost of conversion is aligned with its social cost, so private marginal conditions coincide with the planner's and the decentralized allocation is efficient.

[9] Appendix A.5 provides the formal proof via the planner's Lagrangian.

The two first-best instruments differ in how this correction is implemented. τ_j is a price instrument: a single rate levied at the point of conversion that internalizes the total biodiversity damage associated with production in j. Its incidence is then determined by market conditions. T_{kj} is instead a bilateral transfer from consuming country k to producing country j, with the amount proportional to k's share of j's output. It therefore does not rely on price pass-through to allocate the corrective payment across countries. In this sense, the two instruments implement the same allocation through different institutional arrangements: τ_j decentralizes the first best through prices at the source of conversion, whereas T_{kj} does so through transfers between the countries connected by trade.

The policy implication is that territorial measures need not capture the biodiversity cost of production or consumption when species ranges and supply chains cross borders. As emphasized in the wealth- and footprint-based accounting framework of Dasgupta (2021), accounting for biodiversity impacts therefore requires tracing pressures beyond the territory in which they are realized. In the present model, the omitted component is precisely the set of off-diagonal terms $\phi\, n_i\, \omega_{ij}\, m_j$ for $i \neq j$: exactly what τ_j and T_{kj} are built to price, and what any policy relying on domestic accounting alone will miss.

In sum, the model identifies the channels through which economic sophistication shapes domestic biodiversity. Proposition III determines how structural change reallocates habitat conversion across countries, while Equation (2.4) determines how that reallocation is transmitted through overlapping species ranges. Whether economic sophistication ultimately increases biodiversity threat, and whether the foreign channel accounts for this relationship, are therefore empirical questions. The next section tests these predictions directly.

3 Data and Empirical Strategy

The theoretical framework yields four empirical predictions. First, economic sophistication affects biodiversity through the channels captured by the model, implying a systematic relationship between economic complexity and extinction risk. Second, this relationship should not be explained by domestic land-use or production channels, since the model's mechanism operates through habitat conversion abroad rather than at home. Third, because production and consumption are geographically separated through trade, more sophisticated economies may be net importers of extinction risk. Fourth, because species ranges cross borders, a country's own biodiversity outcomes should respond to habitat conversion embodied in imports from range-overlapping trading partners. The empirical analysis maps directly onto these predictions. We first estimate the within-country relationship between economic sophistication and extinction risk, and test whether domestic land-use and production adjustments account for it. We then test whether economic sophistication is associated with the displacement of biodiversity pressure across borders through trade. Finally, we test whether that displaced pressure feeds back into a country's own biodiversity outcomes through range-overlapping imports. The following section describes the data, followed by the empirical strategies used to evaluate these predictions.

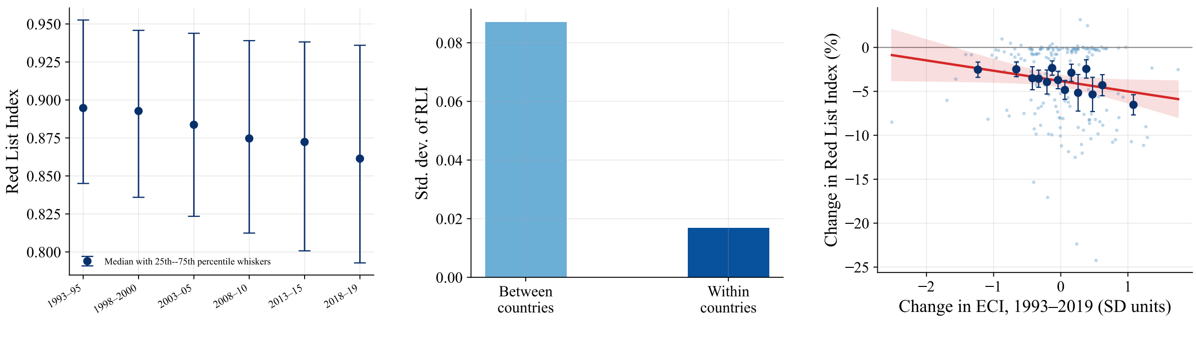
3.1 Data

Extinction risk. The dependent variable is the Red List Index (RLI), which tracks aggregate extinction risk across groups of species (Butchart et al. 2004, 2007, Hoffmann et al. 2010). The index ranges from zero, when all species in a group are extinct, to one, when all are classified as Least Concern; higher values therefore indicate lower extinction risk. It is the empirical counterpart of the biodiversity level b_i. By construction, the RLI changes only when a species is formally reassessed and its IUCN Red List category is revised accordingly (IUCN 2012, BirdLife International & Handbook of the Birds of the World 2019).

Two features of the RLI are central to its interpretation in our setting. First, extinction-risk status is assigned at the species level: a species' Red List category is therefore common to all countries within its geographic range. Second, species ranges frequently cross national borders, so countries share species whose extinction risks are jointly determined by conditions across their ranges. This spatial structure is particularly pronounced for species associated with transboundary ecosystems and for migratory or widely distributed taxa. Consequently, a country's RLI captures the extinction risk of species occurring within its territory even when part of the pressures affecting those species arise outside its borders. In the sample, the RLI averages 0.87, ranging from 0.42 in Mauritius to near unity in Finland and Sweden.

Economic sophistication. The main explanatory variable is the Economic Complexity Index (ECI), constructed by the Growth Lab at Harvard University from countries' export structures (Hidalgo & Hausmann 2009, Hausmann et al. 2007, 2014, Growth Lab at Harvard University 2024). The index is higher for economies that produce a diverse set of sophisticated products that are relatively rare globally, and lower for economies concentrated in a narrower set of ubiquitous, typically resource- or land-intensive products. We standardize the ECI so that coefficients correspond to a one-standard-deviation change.

Because the ECI captures productive capabilities embedded in the overall structure of a country's exports (Hidalgo et al. 2007, Hidalgo 2021), it changes relatively slowly. A short-run shock to a single sector, including one directly exposed to biodiversity conditions such as fisheries or ecotourism, has limited scope to alter the diversity and sophistication of the export basket as a whole. Contemporaneous income, by contrast, can respond immediately to such sector-specific shocks. The ECI is therefore less mechanically exposed than income to short-run sectoral fluctuations that may themselves be related to biodiversity outcomes.



(a) Median RLI by period **(b)** Between vs. within variation of RLI **(c)** ΔECI vs. ΔRLI, 1993–2019

Figure 1: Extinction risk and economic complexity

Notes: Panel (a) plots the median Red List Index across countries in each of the six periods (1993–95, 1998–2000, 2003–05, 2008–10, 2013–15, 2018–19); the whiskers at each period span the 25th to the 75th percentile. Panel (b) decomposes the standard deviation of the Red List Index into its between-country and within-country components. Panel (c) plots, for each country, the change in the Red List Index against the change in the ECI over 1993–2019 (horizontal axis in standard-deviation units; vertical axis the corresponding percentage change in the Red List Index). Each faint point is a country; solid markers are means within equal-count bins of the ECI, with ± 1 standard-error bars, and the shaded band is the 95% confidence interval of the fitted line.

Figure 1 summarizes the empirical structure. Panel (a) shows a persistent decline in the median RLI across the six periods, from 0.89 in 1993–95 to 0.86 in 2018–19, while the interquartile spread widens over the same period. Thus, extinction risk increased over time at the median, alongside a growing dispersion in biodiversity outcomes across countries. Panel (b) documents that within-country variation is substantially smaller than cross-country variation, reflecting the slow reassessment cycle of extinction-risk categories. This motivates a fixed-effects and long-difference strategy, which identifies relationships primarily from medium-run changes. Panel (c) plots changes in RLI against changes in the ECI over 1993–2019; the slope is negative, indicating that countries experiencing increases in the ECI also experienced increases in extinction risk.

Species-range overlap. The theoretical framework's overlap matrix $\Omega = (\omega_{ij})_{1\leq i,j\leq N}$ captures the geographic linkage through which habitat conversion in one country can affect biodiversity outcomes associated with species shared across countries in Equation (2.4). We construct Ω directly from species range polygons for the four taxonomic groups underlying the RLI: mammals, amphibians, and reef-building corals from the IUCN Red List spatial data portal (IUCN 2024), and birds from BirdLife International (BirdLife International & Handbook of the Birds of the World 2025). Let S_i denote the set of assessed species present in country i, pooled across the four groups. Our primary measure is the asymmetric overlap

$$\omega_{ij} \;=\; \frac{|S_i \cap S_j|}{|S_i|}, \tag{3.1}$$

which measures the share of country i's assessed species whose ranges also extend into country j. As a robustness check, we construct a symmetric Sørensen measure,

$$\omega_{ij}^{\text{Sor}} \;=\; \frac{2\,|S_i \cap S_j|}{|S_i| + |S_j|}, \tag{3.2}$$

which weights shared species relative to the combined species richness of the two countries. We also consider a coarser same-continent indicator,

$$\omega_{ij}^{c} \;=\; \mathbf{1}_{\{c(i)=c(j)\}}, \tag{3.3}$$

where $c(\cdot)$ assigns each country to one of seven continents using the country boundaries and geographic classifications in the Natural Earth Admin 0 dataset. Appendix D provides the full data sources, filtering criteria, and construction procedure and reports results under each alternative measure.

Controls and panel structure. Control variables include GDP per capita, population density, and protected-area coverage from the World Bank (The World Bank 2020), and the Human Development Index from UNDP (United Nations Development Programme 2020). The dataset is organized into six observation periods between 1993 and 2019: 1993–95, 1998–2000, 2003–05, 2008–10, 2013–15, and 2018–19. Within each period, we use two- or three-year averages to reduce short-run noise in the economic variables. The five-year spacing between periods is motivated by the biology of the dependent variable: IUCN species are typically reassessed at intervals that vary by taxon, typically four to ten years (IUCN 2012, Butchart et al. 2007), implying limited meaningful annual variation in the RLI. Five-year jumps therefore provide the finest temporal spacing consistent with the reassessment frequency of the underlying biodiversity measure.[10] The panel covers up to 179 countries.

Mechanism variables include land-use shares (agricultural, forest, and urban), agricultural land extent, fertilizer and pesticide use, agricultural productivity and output, and forest-loss indicators, drawn from the World Bank, FAO, and the ESA Climate Change Initiative (CCI) Land Cover dataset. For the leakage analysis, we use a bilateral extinction-risk footprint matrix for 2013, which allocates biodiversity pressure embodied in consumption across country pairs (Irwin et al. 2022, Lenzen et al. 2012, Moran & Kanemoto 2017). From this matrix, we construct domestic, imported, exported, and net extinction-risk footprints for each country.

3.2 Econometric specification

Baseline. We estimate a two-way fixed effects panel specification relating the Red List Index to the ECI:

$$\log RLI_{it} = \beta_0 + \beta_1 ECI_{it} + \beta_2 \log GDP_{it} + \beta_3 \log Popden_{it} + \beta_4 HDI_{it} + \theta_i + \delta_t + \varepsilon_{it}, \tag{3.4}$$

where RLI_{it} is the Red List Index and ECI_{it} is the standardized Economic Complexity Index. The terms θ_i and δ_t denote country and period fixed effects, absorbing all time-invariant country characteristics (including geography, scale, and baseline biodiversity) and global shocks common to all countries.

The control set is intentionally restricted to variables that separate economic structure from scale and development. GDP per capita (GDP_{it}, constant 2017 PPP US dollars) captures the level of economic activity; population density ($Popden_{it}$) proxies direct anthropogenic pressure on habitat (Cincotta et al. 2000); and the Human Development Index (HDI_{it}) captures broader institutional and development correlates of conservation capacity. Additional covariates (including land-use intensity, protected areas, and production structure) are introduced individually in Section 4.2 to assess robustness and mechanisms. Standard errors are clustered at the country level. The coefficient β_1 is therefore identified from within-country variation in the ECI conditional on income and other controls.

We complement the two-way fixed effects specification with a long-difference estimator that exploits medium-run variation in both biodiversity outcomes and the ECI. This approach is motivated by two considerations. First, the Red List Index evolves slowly because species' extinction-risk categories are reassessed only at multi-year intervals, implying limited meaningful high-frequency variation. Second, the theoretical framework emphasizes structural changes in production and their persistent ecological consequences, which are more naturally captured by medium-run shifts in the ECI than by short-run fluctuations. The long-difference specification therefore reduces attenuation arising from measurement noise and provides a robustness check based on lower-frequency variation. The estimator relates changes in extinction risk over the full sample period to corresponding changes in the ECI and the control variables:

$$\Delta \log RLI_i = \beta_1 \Delta ECI_i + \beta_2 \Delta \log GDP_i + \beta_3 \Delta \log Popden_i + \beta_4 \Delta HDI_i + \Delta\varepsilon_i, \tag{3.5}$$

where Δ denotes the change between the first and last period in the panel. This transformation removes

[10] We show that the results are robust to coarser temporal aggregation in a separate robustness check.

all time-invariant country characteristics, including geography, baseline biodiversity endowments, and historical development trajectories. The long-difference estimator can thus be interpreted as a medium-run analogue of the fixed-effects specification: whereas the latter exploits within-country deviations around time means, the former compares initial and final outcomes over a twenty-five-year horizon. Consistency of the two approaches strengthens the interpretation that the relationship between economic sophistication and extinction risk is not driven by short-run noise or transitory shocks, but reflects persistent changes in economic structure.

Instrumental variables. To strengthen the causal interpretation of the relationship between economic complexity and extinction risk, we instrument the Economic Complexity Index (ECI) with the Complexity Outlook Index (COI). The COI measures a country's scope for diversifying into more sophisticated products given its existing position in the global product space (Hausmann et al. 2014, Hidalgo 2021). It therefore provides a source of variation in realized economic complexity that is determined by the country's diversification opportunities rather than by its contemporaneous export outcomes. The first stage is

$$ECI_{it} = \pi_0 + \pi_1 COI_{it} + \pi_2 \log GDP_{it} + \pi_3 \log Popden_{it} + \pi_4 HDI_{it} + \theta_i + \delta_t + u_{it}. \tag{3.6}$$

The second stage replaces ECI_{it} in Equation (3.4) with its fitted value.

The key identifying assumption is the exclusion restriction: conditional on the fixed effects and controls, COI affects extinction risk only through its effect on realized economic complexity. The main concern is that COI is constructed from a country's existing position in the global product space, which reflects decades of past production. Historical production may have left a persistent biodiversity footprint through accumulated habitat conversion, infrastructure, or other changes that continue to affect species' extinction risk today. Because the RLI changes only when species are reassessed, such legacy effects could persist in current biodiversity outcomes even after the underlying production activity has declined. In that case, COI could predict current extinction risk independently of its effect on ECI, violating the exclusion restriction.

Our empirical tests focus directly on this concern. First, we augment the structural equation with COI while controlling for ECI. This specification asks whether the instrument predicts extinction risk conditional on the endogenous variable it instruments. A significant coefficient on COI would indicate a direct channel from the historical product-space position to biodiversity outcomes and would therefore cast doubt on the exclusion restriction. The estimated COI coefficient is small and statistically insignificant, providing no evidence of such a direct effect. Second, we use a lead of COI as a falsification test. The lead does not significantly predict current biodiversity outcomes, providing no evidence that the instrument captures pre-existing movements in extinction risk.

Country and period fixed effects account for time-invariant country characteristics and common global shocks, while the controls for GDP per capita, population density, and HDI account for changes in income, population pressure, and broader development. These adjustments reduce the scope for alternative channels correlated with diversification opportunities. Taken together, the exclusion-restriction tests provide evidence consistent with COI having no independent effect on extinction risk beyond its effect on ECI, while the first-stage relationship separately confirms that COI is a strong predictor of ECI in the first place. Neither the direct-effect test nor the lead-of-COI falsification test can establish the exclusion restriction conclusively, but the direct-effect specification is specifically designed to detect the most consequential legacy channel in this setting and finds no evidence of it.

4 Results

This section presents the paper's four empirical results. We first test the within-country implication of the model linking economic sophistication to extinction risk, assess its causal interpretation using an instrumental-variables strategy, and test whether domestic land-use and production channels explain it. We then investigate the international transmission channel using a bilateral matrix of extinction-risk footprints. Finally, we evaluate the range-overlap mechanism by relating imported biodiversity pressure to changes in domestic extinction risk, and, as a further check, whether biodiversity outcomes are spatially dependent across countries, as the theoretical framework's overlapping species ranges also predict.

4.1 Main results

Table 1 reports the main estimates. Economic sophistication has a negative effect on the Red List Index, implying that more sophisticated economies face greater aggregate extinction risk. The columns add controls sequentially within each design: the ECI alone (columns 1 and 4), the ECI with log GDP per capita (columns 2 and 5), and the full set of controls, additionally including log population density and the Human Development Index (columns 3 and 6). The ECI coefficient is stable across specifications. In the two-way fixed-effects (TWFE) design, a one-standard-deviation increase in ECI lowers the Red List Index by 1.1–1.2 percent (columns 1–3), with virtually no change as controls are added. The long-difference estimates are larger, at 2.1–2.2 percent (columns 4–6), but are similarly stable across specifications.

The coefficient on the Economic Complexity Index is robust across specifications. Adding income per capita, population density, and the Human Development Index (HDI) as controls does not displace it; if anything, the estimate strengthens slightly in magnitude. Sophistication therefore captures a distinct margin and is not standing in for any of these standard development covariates.[11] The magnitudes are modest because the Red List Index moves very little within countries over time. Its within-country standard deviation is roughly one-fifth of its overall standard deviation, since species' extinction-risk categories are reassessed at intervals that vary by taxon, typically four to ten years.

Table 2 reports the instrumental-variables estimates. Instrumenting ECI with COI yields consistently negative and statistically significant estimates across specifications, with the coefficient remaining stable as controls are added. The estimates range from -0.018 to -0.016 and are significant at the 10% level or better.[12] The TWFE coefficient on the Economic Complexity Index is -0.012. All three instrumented

[11] Appendix C.4 makes this explicit with two additional specifications. A residualized-ECI variant projects the ECI onto log GDP per capita, log population density, and the Human Development Index (plus country and period fixed effects); the coefficient on the residualized ECI equals the baseline estimate by the Frisch–Waugh–Lovell theorem. An interaction between the ECI and mean-centred log GDP is statistically indistinguishable from zero ($p = 0.78$), so the sophistication effect does not vary with income. The within-country correlation between the ECI and log GDP per capita is $r = 0.06$.

[12] The COI is a strong, valid instrument: first-stage $F = 31.4$; a direct test of the exclusion restriction and a falsification test using its lead are both statistically insignificant. Appendix B reports the full battery of instrument-validity tests.

Table 1: Baseline estimates of the effect of economic sophistication on extinction risk

	TWFE			Long difference		
	(1)	(2)	(3)	(4)	(5)	(6)
Economic Complexity Index	-0.011***	-0.011***	-0.012***	-0.022***	-0.021***	-0.022***
	(0.004)	(0.004)	(0.004)	(0.007)	(0.007)	(0.006)
log GDP per capita		-0.003	-0.020**		-0.011	-0.047***
		(0.007)	(0.008)		(0.010)	(0.012)
log Population density			-0.030***			-0.054***
			(0.011)			(0.013)
Human Development Index			0.134**			0.286***
			(0.061)			(0.106)
Country FE	Yes	Yes	Yes	No	No	No
Period FE	Yes	Yes	Yes	No	No	No
Within R^2	0.396	0.397	0.418	0.056	0.064	0.141
Observations	1,000	1,000	1,000	129	129	129
Countries	179	179	179	129	129	129

Notes: This table estimates the within-country TWFE model, Equation (3.4) (columns 1–3), and its long difference between the first and last periods, Equation (3.5) (columns 4–6). Columns add controls sequentially: ECI alone (columns 1 and 4); ECI and log GDP per capita (columns 2 and 5); the full set of controls (columns 3 and 6). The dependent variable is the log Red List Index. The Economic Complexity Index is standardized, so its coefficient is the effect of a one-standard-deviation change. Standard errors in parentheses are cluster-robust by country (columns 1–3) and heteroskedasticity-robust (columns 4–6). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 2: Instrumental-variables estimates

	IV			TWFE
	(1)	(2)	(3)	(4)
Economic Complexity Index	-0.018*	-0.018*	-0.016*	-0.012***
	(0.010)	(0.010)	(0.009)	(0.004)
log GDP per capita		-0.003	-0.020**	-0.020**
		(0.006)	(0.008)	(0.008)
log Population density			-0.031***	-0.030***
			(0.011)	(0.011)
Human Development Index			0.143**	0.134**
			(0.058)	(0.061)
First-stage F	33.4	32.9	31.4	
Observations	1,000	1,000	1,000	1,000
Countries	179	179	179	179

Notes: This table shows estimates using the six-period (5-year-window) panel, 1993–2019, with country and period fixed effects. Columns 1–3 instrument the standardized Economic Complexity Index with the contemporaneous Complexity Outlook Index (Hausmann et al. 2014, Hidalgo 2021), adding controls sequentially: column 1 has no additional controls, column 2 adds log GDP per capita, and column 3 adds the full control set. Column 4 reports the TWFE estimate with the full control set of Equation (3.4) (log GDP per capita, log population density, HDI). Standard errors clustered by country in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

estimates are negative and comparable in magnitude to the TWFE estimate. Although instrumental-variables estimators are less efficient than TWFE and therefore produce wider confidence intervals, the fully controlled IV specification remains statistically indistinguishable from the TWFE estimate. Correcting for potential endogeneity therefore does not alter either the sign or the economic magnitude of the relationship between economic sophistication and extinction risk.

Taken together, the TWFE, long-difference, and IV estimates converge on a robust causal finding: greater economic sophistication increases a country's aggregate extinction risk. The direction of this within-country result is harder to reconcile with the cross-sectional biodiversity–income literature, which finds an inverted-U between income and species threat and suggests that further growth may ease pressure on biodiversity (Naidoo & Adamowicz 2001, Asafu-Adjaye 2003, Dietz & Adger 2003, Mills & Waite 2009); it also differs in direction from the complexity-and-carbon literature, which finds that complexity lowers territorial emissions (Romero & Gramkow 2021, Mealy & Teytelboym 2022, Boleti et al. 2021). The closest precedent in our direction is Neagu (2020), who finds that complexity raises the aggregate ecological footprint across 48 complex economies. Our panel evidence is complementary: it isolates the effect at the biodiversity margin and uses within-country variation, addressing the cross-country confounding that has long been a concern in this literature (Stern 2004).[13]

4.2 Additional robustness and domestic channel tests

The baseline relationship between economic sophistication and extinction risk is robust to the checks that matter most in a short within-country panel. This subsection also examines the domestic mechanisms that could in principle generate this relationship and shows that they do not account for the estimated effect.

Sensitivity to controls. A natural concern is that economic sophistication proxies for omitted domestic determinants of habitat loss. We address this by augmenting the preferred specification, which already includes income, population density, and the Human Development Index, with additional controls one at a time. Each variable is selected on theoretical grounds rather than for exhaustive conditioning. We consider conservation effort (protected-area coverage), land-use composition (forest, agricultural, and urban shares), production intensity (energy use, fertilizer and pesticide application, and agricultural productivity), and sectoral structure (manufacturing, services, and agricultural value added).

Table 3 reports the corresponding estimates. The coefficient on the ECI is highly stable, remaining between -0.008 and -0.013 and statistically significant in all specifications. No measure of domestic land use, agricultural intensity, or sectoral composition meaningfully attenuates the effect. Agricultural inputs and productivity variables produce modest attenuation in smaller samples, but the coefficient remains negative and significant throughout. Likewise, conditioning on the sectoral composition of value added leaves the estimate essentially unchanged, indicating that the ECI is not proxying for the dominance of specific sectors but rather for the sophistication of production.

These results clarify two distinct questions. First, domestic land use is an important determinant of biodiversity outcomes: forest cover, the most direct measure of habitat availability, enters positively and significantly ($p < 0.05$), implying that countries with greater forest endowment exhibit lower extinction

[13] Appendix C.3 reports between-country, pooled-OLS, and within-country fixed-effects estimates side by side, with an EKC (quadratic-income) variant of each. The Economic Complexity Index coefficient is positive in cross-section ($+0.048$, $p < 0.01$, Table C.4 column 1) and flips to negative within countries (-0.012, $p < 0.01$, column 5), the empirical signature of the cross-sectional confounding cautioned by Stern (2004).

Table 3: Stability of the ECI coefficient under additional controls

	(1) Pref.	(2) +Prot. area	(3) +FDI	(4) +Forest	(5) +Cropland	(6) +Urban	(7) +Energy	(8) +Fert.	(9) +Pest.	(10) +AgTFP	(11) +Manuf.	(12) +Serv.	(13) +Agric.
Economic Complexity Index	-0.012***	-0.010**	-0.012***	-0.012***	-0.012***	-0.012***	-0.010***	-0.008**	-0.009**	-0.010**	-0.013***	-0.012***	-0.012***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Protected-area share		0.134*											
		(0.073)											
FDI (% of GDP)			-0.000										
			(0.000)										
Forest share				0.287**									
				(0.140)									
Agricultural land share					0.000								
					(0.000)								
Urban land share						-0.099							
						(0.061)							
log Energy consumption							-0.012**						
							(0.005)						
log Fertilizer (N)								0.004*					
								(0.002)					
log Pesticides									0.002				
									(0.003)				
Agricultural TFP (log)										-0.000			
										(0.007)			
log Manufacturing VA share											0.006		
											(0.004)		
log Services VA share												0.008	
												(0.007)	
log Agriculture VA share													-0.001
													(0.005)
Within R^2	0.418	0.408	0.420	0.444	0.419	0.418	0.448	0.432	0.427	0.415	0.416	0.392	0.418
Observations	1,000	810	997	993	995	993	992	758	718	920	930	946	970

Notes: Column 1 reports the preferred within-country specification, Equation (3.4) (country and period fixed effects, log income per capita, log population density, and HDI). Columns 2–13 add, one at a time, the additional control named at the head of the column; only the coefficient on the standardized Economic Complexity Index (row 1) and on the added control (its own row) is shown, with the base controls suppressed for compactness. Units: protected-area share, forest share, agricultural land share, and urban land share are percent of land area; FDI, manufacturing, services, and agriculture value-added shares are percent of GDP; energy consumption is kg of oil equivalent per capita; fertilizer (N) and pesticides are metric tons; agricultural TFP is an index. Standard errors clustered by country in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

risk. Second, however, the effect of economic sophistication does not appear to operate through domestic land-use margins.

Sample, functional form, and frequency. Three additional robustness checks, reported in Appendix C, confirm the stability of the baseline estimate. First, restricting the sample to the balanced panel of countries observed in all six periods leaves the coefficient essentially unchanged at -0.013 ($p < 0.01$). Second, allowing for a quadratic in income does not generate evidence of a biodiversity Environmental Kuznets Curve: neither the linear nor the quadratic income term is statistically significant, and the coefficient on the ECI remains stable, consistent with the view that the EKC pattern in cross-country data primarily reflects income-level differences rather than within-country dynamics (Stern 2004, Vincent 1997). Third, re-aggregating the data into three longer periods (1993–95, 2005–07, and 2017–19) yields a slightly larger estimate of -0.015 ($p < 0.01$), indicating that the result is not sensitive to the temporal frequency of observation.

Across all specifications, the coefficient on the ECI remains tightly clustered around its preferred estimate of -0.012 and retains statistical significance, indicating that the baseline relationship is robust to alternative sample restrictions, functional form assumptions, and temporal aggregation.

Reverse causality. Another concern is whether economic sophistication itself could be responding to biodiversity decline rather than causing it, for instance if the contraction of biodiversity-dependent sectors pushed a country toward more sophisticated activities. Reverse causality, however, is unlikely to explain the estimated relationship between economic complexity and biodiversity outcomes. Economic complexity measures the accumulation of productive capabilities and reflects long-run patterns of structural transformation, technological knowledge, and export diversification. By contrast, biodiversity loss affects economic activity primarily through ecosystem-service channels such as pollination, water regulation, and soil fertility, which have only an indirect connection to the productive capabilities underlying economic complexity. While biodiversity degradation may influence economic performance over long horizons, there is little theoretical basis to expect changes in species extinction risk to systematically alter export sophistication in the short or medium run. The more plausible direction of causality runs from economic complexity to biodiversity outcomes through changes in production structures, resource demand, and trade linkages.

A related concern is that biodiversity decline may increase environmental awareness and public demand for conservation policies or greener production practices. If so, environmental awareness would arise as a response to biodiversity deterioration rather than as an independent determinant of biodiversity outcomes. Such a mechanism would tend to mitigate biodiversity losses and improve Red List Index outcomes, thereby attenuating rather than generating the negative relationship between economic complexity and biodiversity outcomes. More generally, one might worry that broader development processes simultaneously promote both economic sophistication and environmental protection. To account for this possibility, the regressions control for the Human Development Index (HDI), which captures important dimensions of socioeconomic development associated with environmental preferences and conservation capacity. Consistent with this interpretation, HDI enters positively in the regressions, indicating that more developed societies tend to achieve better biodiversity outcomes. Yet the estimated coefficient on economic complexity remains negative and significant after controlling for HDI. This suggests that the relationship between economic complexity and biodiversity outcomes is unlikely to be driven solely by environmental awareness or development-related conservation efforts.

First-link tests for the domestic channel. We next examine eleven candidate domestic channels through which economic sophistication could affect biodiversity by changing demand for land-intensive inputs, production intensity, or sectoral composition. For each channel, we regress the candidate outcome on standardized ECI with country and period fixed effects and the baseline controls. Ten of the eleven coefficients are statistically indistinguishable from zero at conventional levels. The only detectable relationship is with the manufacturing value-added share, which increases modestly with sophistication (0.08, $p = 0.07$), indicating a shift toward manufacturing and away from more land-intensive activities. This compositional adjustment is informative for the interpretation of the main result: economic sophistication is accompanied by domestic structural transformation, but this transformation does not account for the estimated biodiversity effect. Table 3 shows that the ECI coefficient is essentially unchanged when the manufacturing share is added to the baseline specification. Appendix C.2 reports the complete set of channel estimates.

The absence of evidence for domestic channels points to a different interpretation of the main result. The biodiversity cost associated with greater sophistication may not be borne primarily within the country that becomes more sophisticated. As the next section shows, more sophisticated economies increasingly source their consumption footprint abroad, potentially displacing biodiversity pressure through trade. The resulting mechanism is therefore trade-embodied rather than territorial: domestic land-use accounts need not capture the biodiversity pressure associated with higher economic sophistication. This interpretation contrasts with mechanism-based accounts of the biodiversity–income relationship that attribute improvements in environmental outcomes to greater domestic land-use efficiency or stronger conservation institutions. Instead, it is consistent with the pollution-haven perspective (Copeland & Taylor 1994, 2004), examined empirically by Cole (2004), in which more sophisticated economies may relocate ecologically intensive production toward trading partners rather than eliminate the underlying environmental pressure.

4.3 The geography of the extinction-risk footprint

As the results from the previous section showed, agricultural expansion, forest loss, input intensity, and agricultural productivity do not respond systematically to economic sophistication, and conditioning on these variables does not alter the baseline relationship between sophistication and extinction risk. This pattern indicates that domestic land-use adjustments are not the main transmission margin.

The implication is that the biodiversity cost of economic sophistication is geographically displaced. Rather than being eliminated, extinction pressure is embodied in production networks and reallocated across countries through trade. In supplier economies, this displacement is associated with land conversion for export-oriented agriculture (including plantations and aquaculture), extraction activities integrated into global value chains, and the associated chemical and infrastructural pressures (fertilizer and pesticide runoff, mining impacts, and transport infrastructure). The extinction-risk footprint framework (Irwin et al. 2022, Lenzen et al. 2012, Moran & Kanemoto 2017) aggregates these channels into a bilateral accounting system that assigns biodiversity pressure embodied in consumption to its ultimate geographic origin.

We examine this mechanism using the 2013 cross-sectional matrix of extinction-risk footprints:[14]

$$Y_i = \alpha + \delta_1 ECI_i + \delta_2 \log GNI_i + u_i, \tag{4.1}$$

where Y_i is, in turn, the log imported footprint, the log domestic footprint, the foreign share of the total footprint, and the net import position (Table 4); $\log GNI_i$ is the log of Gross National Income per capita. We use GNI rather than GDP here because the outcomes are consumption-based: GNI includes net factor income from abroad and therefore better captures consumption capacity than territorial GDP, which is the convention in the multi-regional input–output literature (Lenzen et al. 2012, Moran & Kanemoto 2017, Irwin et al. 2022). This specification provides the empirical counterpart to the model prediction in Proposition III, which predicts that both the foreign share of the consumption footprint and net imported extinction risk increase with economic sophistication. The test isolates whether economically more sophisticated countries systematically differ in the geography of the biodiversity pressure associated with their consumption.[15]

Table 4: Economic sophistication, income, and the geography of the extinction-risk footprint

	(1) log Imported	(2) log Domestic	(3) Foreign share	(4) Net import
Economic Complexity Index	0.487***	-0.255	0.123***	633.601***
	(0.162)	(0.246)	(0.024)	(222.164)
log GNI per capita	0.545***	-0.730***	0.125***	208.278
	(0.132)	(0.235)	(0.023)	(161.467)
R^2	0.326	0.108	0.474	0.154
Observations	173	173	173	173

Notes: Each column estimates the 2013 cross-sectional regression Equation (4.1) for a different outcome: the log imported footprint (1), the log domestic footprint (2), the share of the consumption footprint falling abroad (3), and the net import position (4). "Net import" is imported minus exported extinction-risk footprint, positive for net importers. The Economic Complexity Index is standardized. Heteroskedasticity-robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Two results are central. First, the geographic composition of the biodiversity footprint shifts sharply with economic sophistication. The foreign share of a country's consumption footprint increases strongly in the ECI: a one-standard-deviation rise in the ECI raises this share by approximately twelve percentage points, and together with income it explains close to half of its cross-country variation (column 3). Second, the net position reverses with sophistication: while less sophisticated economies are net exporters of extinction risk, more sophisticated economies are net importers (column 4).

This pattern is reflected in both margins of the decomposition. More sophisticated economies are associated with a larger imported footprint (column 1), while domestic footprint declines with income

[14] The bilateral extinction-risk footprint matrix of Irwin et al. (2022) is currently available only for 2013; a panel version has not yet been constructed.

[15] The Economic Complexity Index and the extinction-risk footprint both ultimately trace to international trade-flow statistics (ECI via UN Comtrade, the footprint via the 2013 Eora multi-region input–output table, itself calibrated from trade data), raising the possibility that the relationship partly reflects gross trade volume rather than sophistication specifically. Appendix C.5 re-estimates Equation (4.1) controlling for trade openness (World Bank, trade as a share of GDP, 2013) (The World Bank 2020); the Economic Complexity Index coefficient is essentially unchanged, or strengthens, on all three outcomes.

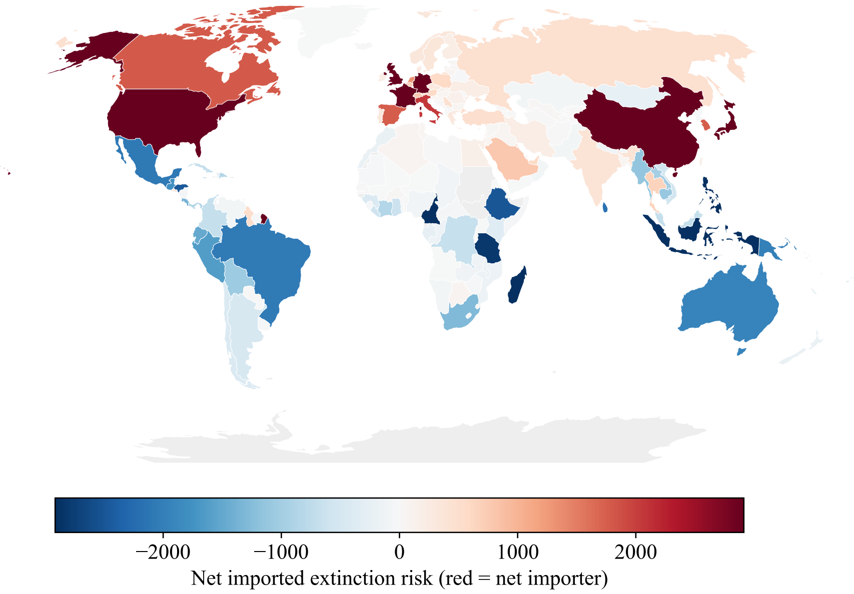


Figure 2: The geography of outsourced extinction risk, 2013.

Notes: Each country is shaded by its net imported extinction risk (the extinction risk its consumption causes abroad minus the risk foreign consumption causes within it). Red denotes net importers (consumption causes more loss abroad than is borne at home) and blue denotes net exporters. Values are computed from the 2013 bilateral extinction-risk footprint matrix (Irwin et al. 2022, Lenzen et al. 2012).

(column 2). The joint implication is a reallocation of biodiversity pressure across space: high-income, high-sophistication economies generate relatively limited domestic extinction risk but increasingly rely on external ecosystems to sustain consumption.

The gradient is economically large across the distribution. Countries in the top tercile of sophistication place roughly seventy percent of their consumption footprint abroad and are net importers of extinction risk, whereas those in the bottom tercile externalize about twenty percent and are net exporters. Figure 2 summarizes the resulting geography: North America, Western Europe, and East Asia are systematic net importers, while Latin America, sub-Saharan Africa, and Southeast Asia are net exporters. This spatial pattern complements the within-country results: although biodiversity risk rises with sophistication domestically, an increasing share of the associated pressure is displaced onto trading partners' ecosystems.

This evidence aligns with the multi-regional input–output literature documenting the displacement of biodiversity and ecological pressure from high-income to biodiversity-rich economies (Lenzen et al. 2012, Moran & Kanemoto 2017, Marques et al. 2019, Wilting et al. 2017, Chaudhary & Kastner 2016, Irwin et al. 2022). The contribution here is to show that this displacement is not explained by income alone: even conditional on income, economic sophistication is a key structural determinant of the foreign share of the extinction-risk footprint.

The leakage gradient is continuous across the entire distribution of economic sophistication rather than being driven by a discrete separation between less and more sophisticated economies. Figure 3 plots, for each country in the 2013 cross-section, the share of its consumption footprint embodied abroad against its standardized Economic Complexity Index. The relationship is steep and tightly estimated: a one-standard-deviation increase in the ECI is associated with an increase in the foreign share by approx-

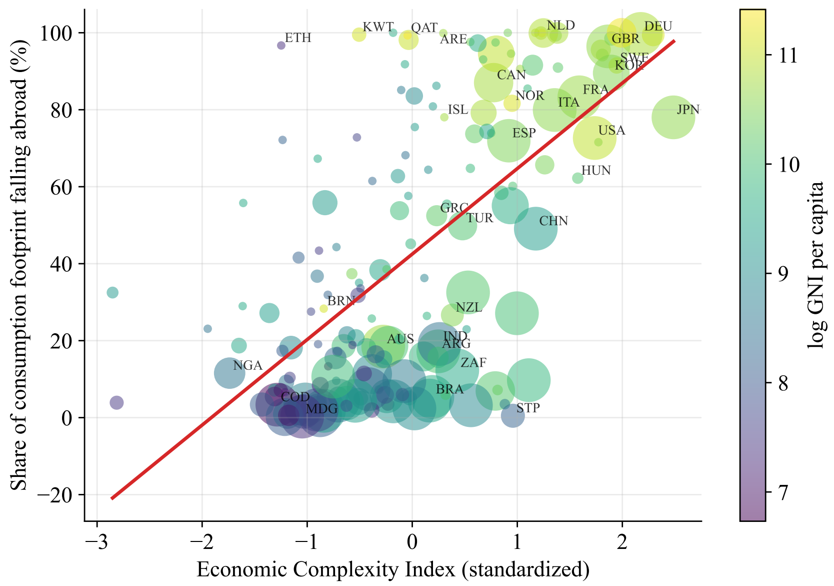


Figure 3: Economic sophistication and the share of the consumption footprint that falls abroad

Notes: Each point is a country in the 2013 cross-section. The vertical axis is the share of a country's consumption footprint that falls on other countries' species; the horizontal axis is the standardized Economic Complexity Index. Point size is proportional to the total consumption footprint and color encodes log GNI per capita. The fitted line corresponds to the foreign-share regression, Equation (4.1).

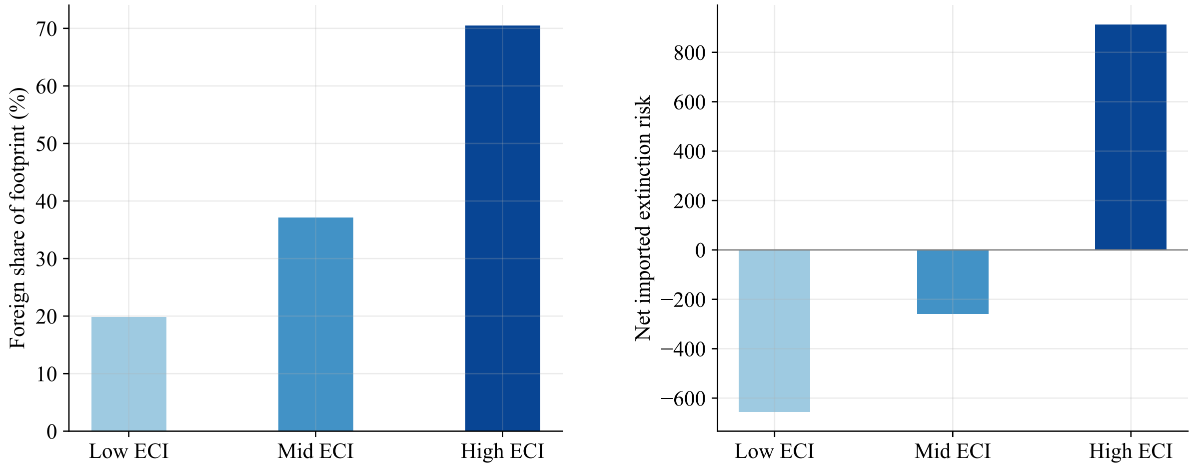
imately twelve percentage points (Table 4, column 3), and together with income it accounts for nearly half of the cross-country variation in this outcome.

The pattern is primarily organized along the sophistication dimension rather than income. Sophistication therefore captures a dimension of production structure that is distinct from income and more closely aligned with the geography of embodied biodiversity pressure. Aggregating the data into terciles yields a consistent and more transparent summary. Figure 4, panel (a), reports mean foreign shares by sophistication group. Even the bottom tercile externalizes roughly twenty percent of its consumption footprint, but this share rises monotonically across the distribution, reaching close to seventy percent in the top tercile. Panel (b) reports the corresponding net position in extinction-risk footprints. Less sophisticated economies are net exporters of biodiversity pressure, whereas more sophisticated economies are net importers.

Taken together, the two panels restate the regression evidence in Table 4 in distributional form. As economic sophistication increases, a growing share of consumption-related biodiversity loss is displaced abroad, and the role of countries shifts from net suppliers to net absorbers of global extinction risk.

4.4 Imports from species-range-overlapping countries and own-country decline

The previous two results establish complementary facts. Within countries, greater economic sophistication is associated with a decline in the Red List Index. Across countries, more sophisticated economies externalize an increasing share of their biodiversity footprint through trade. The theoretical framework links these findings through the species-range overlap matrix Ω. From Equation (2.12), the effect of a country's sophistication on its own biodiversity status operates not only through domestic habitat conversion but also through the foreign component $-\sum_{j\neq i}\omega_{ij}m_j(dh_j/d\kappa_i)$, which captures habitat conversion induced abroad by the country's own consumption and transmitted back through shared species



(a) Share of footprint falling abroad

(b) Net imported extinction risk

Figure 4: The extinction-risk footprint by tercile of economic sophistication

Notes: Countries are grouped into terciles of the standardized Economic Complexity Index. Panel (a) shows the mean share of the consumption footprint that falls abroad; panel (b) shows the mean net imported extinction risk (positive values denote net importers). Both rise monotonically with sophistication.

ranges. The empirical implication is that countries importing a larger biodiversity footprint from range-overlapping regions should experience a larger subsequent decline in their own Red List Index. We test this prediction directly. The single coefficient that identifies the mechanism is on the overlap share,

$$\sigma_i \;=\; F_i^{\omega} \,/\, F_i^{\text{imp}}, \quad \text{with} \quad F_i^{\omega} = \sum_{j \neq i} \omega_{ij} F_{j \to i} \quad \text{and} \quad F_i^{\text{imp}} = \sum_{j \neq i} F_{j \to i}, \tag{4.2}$$

where $F_{j \to i}$ is the extinction-risk footprint generated in source country j by consumption in country i. Unlike the raw imported or ω-weighted footprint, both of which scale with the sheer volume of a country's trade, σ_i holds import scale fixed and isolates the claim distinctive to this theory: that the composition of a country's imports, range-overlapping or not, predicts its own biodiversity decline, not merely their scale. We estimate

$$\Delta \log RLI_i = \alpha + \beta \sigma_i + \Delta \mathbf{z}_i' \gamma + \varepsilon_i, \tag{4.3}$$

where Δ denotes the long difference between the 1993–95 and 2018–19 periods.[16] $\Delta \mathbf{z}_i$ is a control vector: log GDP per capita, log population density, and the Human Development Index in every column, plus $\log F_i^{\text{imp}}$ (import scale) and, in the domestic-channel-controlled columns, $\log F_i^{\text{dom}}$ and $\log F_i^{\text{exp}}$.[17] These scale and domestic-channel terms are controls, not coefficients of independent interest, so Table 5 does

[16]The bilateral footprint matrix is measured only in 2013, the midpoint of this 25-year window, so it is implicitly treated as a stable stand-in for the trade structure driving biodiversity pressure throughout the period. Appendix D.3 reports the same specification, on the identical country samples, restricted to the post-2013 long difference (2013–15 to 2018–19), so the footprint no longer predates any of what it explains; the overlap-share coefficient retains its sign and remains significant under all three Ω measures compared in Table 5.

[17]Using the bilateral extinction-risk footprint matrix $M[i, j]$ defined in Appendix D.2 (the extinction risk borne by source i's species on account of consumption in destination j), $F_i^{\text{dom}} = M[i,i]$ is the domestic footprint, the risk borne by country i's own species on account of its own consumption, and $F_i^{\text{exp}} = \sum_{j \neq i} M[i, j]$ is the exported footprint, the risk borne by i's species on account of consumption elsewhere.

not report them individually; their role is discussed below the table. The theoretical prediction is $\beta < 0$.

Table 5: Overlap share and own-country extinction-risk decline

	(1) Primary	(2) + Domestic	(3) Sørensen	(4) Continent
Overlap share of imports (σ)	-0.265**	-0.410**	-0.333*	-0.180**
	(0.122)	(0.206)	(0.181)	(0.077)
Δ ECI	-0.016***	-0.015***	-0.016***	-0.014***
	(0.005)	(0.005)	(0.005)	(0.005)
ω measure	Species range	Species range	Sørensen	Continent
Domestic-channel control	No	Yes	Yes	Yes
R^2	0.306	0.325	0.284	0.367
Observations	127	127	127	120

Notes: The dependent variable is the long difference in $\log RLI$ between the 1993–95 and 2018–19 windows. σ is the overlap share of imported extinction-risk footprint, constructed from the 2013 bilateral footprint matrix (Irwin et al. 2022) and the overlap matrix $\Omega = (\omega_{ij})_{1\leq i,j\leq N}$ (Appendix D.1). All specifications include log GDP per capita, log population density, the Human Development Index, and log imported footprint. Columns 2–4 additionally control for log domestic and exported footprint and vary the measure of Ω: the primary asymmetric species-range measure (column 2), the symmetric Sørensen measure (column 3), and the same-continent indicator (column 4). Heteroskedasticity-robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5 reports the overlap-share coefficient across four specifications that hold the estimating equation fixed and vary only the robustness dimension, so that a single row tells the whole story. Column 1 is the baseline: σ_i enters negatively and significantly (-0.265, $p < 0.05$), controlling for import scale alone. Column 2 adds the domestic-channel control, $\log F_i^{\mathrm{dom}}$ and $\log F_i^{\mathrm{exp}}$, which together with F_i^{imp} span the demand-source-agnostic damage done to country i's own territory, the empirical counterpart of the domestic term in Equation (2.12). This is the toughest test: does the range-overlap channel survive net of the domestic channel? It does, and the coefficient strengthens (-0.410, $p < 0.05$). Holding this domestic-channel control fixed, column 3 substitutes the symmetric Sørensen overlap measure for the primary asymmetric one: the coefficient remains negative and significant (-0.333, $p < 0.10$). Column 4 substitutes the coarse same-continent indicator: smaller in magnitude (-0.180, $p < 0.05$) but still significant. The ordering across columns 2–4, species-range asymmetric (-0.410) and Sørensen (-0.333) both exceeding the continent proxy (-0.180), is itself evidence for the mechanism: a proxy built from actual species occurrence outperforms a proxy built from geographic classification alone.

Across all four columns the coefficient on the ECI remains stable and significant at the one-percent level (between -0.014 and -0.016), unaffected by which bridge specification is used. The bridge variables therefore contribute explanatory power beyond the ECI itself, providing direct empirical support for the species-range overlap mechanism developed in the theoretical framework. Appendix D describes the construction of Ω and the bridge variables in full.

Spatial dependence. The within-country relationship between economic sophistication and extinction risk could in principle reflect spatial confounding rather than the mechanism proposed here: if neighboring countries have correlated biodiversity outcomes, some unobserved regional factor correlated with both the ECI and the RLI could generate a spurious association. We address this directly by modeling spatial dependence explicitly, rather than assuming it away.

Proposition I predicts that biodiversity levels are positively spatially dependent because species' ranges extend across national borders, and the data strongly support this prediction. Extinction risk clusters geographically: the contiguity Moran's I of the Red List Index is 0.54,[18] and the spatial autoregressive coefficient is positive and highly significant, consistent with the fact that neighboring countries often share ecosystems and that a species classified as threatened carries the same conservation status throughout its range (Pandit & Laband 2007). Table 6 shows that this spatial dependence, real as it is, does not account for the estimated relationship between economic sophistication and extinction risk. We estimate the fixed-effects spatial autoregressive panel specification

$$\log RLI_{it} = \rho \sum_j W_{ij} \log RLI_{jt} + \beta_1 ECI_{it} + \mathbf{z}_{it}'\gamma + \theta_i + \delta_t + \varepsilon_{it}, \tag{4.4}$$

where W_{ij} denotes a row-standardized spatial weight matrix,[19] ρ measures spatial dependence, and θ_i, δ_t are country and period fixed effects, by maximum likelihood (Anselin 1988), on the full six-period panel. This nests the country and period fixed effects of the preferred within-country specification, Equation (3.4), while additionally estimating the spatial-lag term explicitly, so a single model does both jobs: absorbing common shocks and testing the specific spatial-dependence mechanism of Proposition I. Appendix A.4 shows that this linear form follows directly from Proposition I's covariance property, via the best linear projection of b_i onto its spatial lag.

The primary test uses the three species-range overlap measures: asymmetric (primary), symmetric Sørensen, and same-continent (columns 1–3 of Table 6). We then check, as an explicit additional step rather than the starting point, whether the result also survives under queen contiguity (column 4), the geographic weight matrix that is the default choice in the applied spatial-econometrics literature. Maximum likelihood, rather than spatial two-stage least squares, is used throughout: the Ω-based matrices are dense (the vast majority of country pairs share at least one species across the four pooled taxa), and a dense W produces a degenerate spatial-lag instrument under 2SLS, whereas maximum likelihood does not depend on constructing such an instrument and remains well behaved.

The coefficient on the ECI is negative and highly significant, though smaller in magnitude than the non-spatial baseline, under every definition of spatial proximity considered. Under the theoretically motivated species-range measures, it is -0.006 ($p < 0.01$) for both the asymmetric measure (column 1) and the Sørensen alternative (column 2), and -0.009 ($p < 0.001$) under the same-continent construction (column 3). Checked afterward against the geographic default, queen contiguity, the coefficient remains negative and highly significant (-0.007, $p < 0.001$; column 4): the result does not depend on using a biologically grounded W rather than the field's standard geographic one. The spatial autoregressive parameter is positive and highly significant throughout, confirming that biodiversity outcomes are strongly interconnected across countries under every notion of proximity tested; it is markedly larger for the Ω-based species-range matrices ($\rho \approx 0.92$, columns 1–2) than for the continent indicator or queen contiguity ($\rho \approx 0.43$–0.52, columns 3–4), reflecting the much denser connectivity of species-

[18] Moran's I measures spatial autocorrelation: whether values observed at neighboring locations tend to be similar (positive I), dissimilar (negative I), or unrelated (zero), relative to what random spatial variation would produce. Here, "neighboring" means countries that share a land border. I is bounded roughly between -1 and 1, so a value of 0.54 indicates substantial positive clustering: bordering countries have systematically similar Red List Index levels.

[19] Row-standardized means $W_{ij} = \omega_{ij} / \sum_{k \neq i} \omega_{ik}$ for $j \neq i$ and $W_{ii} = 0$, so each row of W sums to one and $(W\mathbf{b})_i$ is a weighted average of country i's neighbors (Appendix A.4).

Table 6: Spatial robustness of the sophistication–extinction-risk association

	(1) Ω asymmetric	(2) Ω Sørensen	(3) Ω continent	(4) Queen contiguity
Economic Complexity Index	-0.006^{***}	-0.006^{***}	-0.009^{***}	-0.007^{***}
	(0.002)	(0.002)	(0.002)	(0.002)
Spatial lag ρ	0.918^{***}	0.911^{***}	0.517^{***}	0.429^{***}
	(0.032)	(0.034)	(0.052)	(0.033)
Countries	119	119	119	119
Observations	714	714	714	714

Notes: The dependent variable is the log Red List Index. Each column estimates the fixed-effects spatial autoregressive panel model, Equation (4.4), by maximum likelihood on the full six-period balanced panel (119 countries $\times$ 6 periods), with country and period fixed effects throughout and W varying by column: the three Ω measures of Appendix D.1 (columns 1–3, used unsparsified and row-standardized), the theoretically primary choice, followed by queen contiguity (column 4, row-standardized; island countries with no touching neighbour are connected to their three nearest neighbours by great-circle distance), the standard geographic baseline added as a further check. The cross-sectional contiguity Moran's I of the Red List Index in the final period (2018–19) is 0.54. All specifications include log GDP per capita, log population density, and the Human Development Index (coefficients omitted). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

range overlap relative to geographic classification, not instability, since it remains below the boundary of one required for a stable spatial process. The evidence therefore supports the final prediction of the theoretical framework: biodiversity loss is spatially transmitted across countries, under both the biologically grounded and the geographic definitions of proximity, yet this spatial dependence does not explain away the within-country relationship between economic sophistication and extinction risk. Taken together, the results support a coherent interpretation. Economic sophistication increases extinction risk within countries, not because it intensifies domestic habitat conversion, but because it increasingly displaces biodiversity pressure through international trade, where it feeds back through shared species ranges.

5 Discussion

The evidence points to a common conclusion: what looks like a biodiversity dividend from economic sophistication is largely an artifact of territorial accounting, since sophistication redistributes extinction risk across borders rather than eliminating it. More sophisticated economies experience rising extinction risk within countries even though this relationship is not explained by domestic land use. Instead, it increasingly shifts the biodiversity footprint of consumption toward foreign ecosystems, where the resulting habitat conversion feeds back through species whose ranges cross national borders. The central implication is that sophistication changes where a country's consumption converts habitat, not whether that conversion happens at all.

This perspective helps reconcile several strands of the literature. Cross-sectional studies of biodiversity and development often report an Environmental Kuznets Curve, suggesting that biodiversity pressure eventually declines with income (Naidoo & Adamowicz 2001, Asafu-Adjaye 2003, Dietz & Adger 2003, Mills & Waite 2009). An early and influential skeptical voice within this literature is Dietz & Adger (2003), who find no robust inverted-U for biodiversity and argue that intentional conservation effort, rather than rising income alone, is the binding margin for protecting species. Mills & Waite (2009) reach a similar skeptical conclusion using deforestation rates as a biodiversity proxy across 35 tropical countries, finding that the EKC pattern is fragile to quantile-regression and spatial-filtering specifications, and that wealth is not a reliable indicator of improved conservation practice. Our within-country evidence is less supportive of an EKC interpretation and instead echoes Dietz & Adger (2003), Mills & Waite (2009), and the caution raised by Stern (2004) for environmental Kuznets curves more generally: cross-country comparisons can confound genuine environmental improvement with the spatial relocation of environmental pressure. The same confounding applies to economic sophistication itself. The ECI coefficient is positive across countries, where more sophisticated economies appear to have lower extinction risk, but negative within countries over time, where rising sophistication goes hand in hand with rising threat. The displacement mechanism that Mills & Waite (2009) anticipate verbally is exactly what we identify and measure.

Our results also complement the economic complexity literature. Whereas complexity has repeatedly been associated with lower territorial carbon emissions and cleaner production (Romero & Gramkow 2021, Mealy & Teytelboym 2022, Stojkoski et al. 2023, Boleti et al. 2021), the closest precedent in our direction is Neagu (2020), who finds that complexity raises the aggregate ecological footprint across complex economies. Biodiversity follows a different logic because habitat conversion is spatially localized and embodied in traded goods. A country may successfully decarbonize production at home while continuing to drive habitat conversion in biodiversity-rich trading partners. Our findings therefore do not contradict the complexity literature; rather, they show that biodiversity responds differently from pollutants whose damages are global and whose accounting remains largely production based.

The biodiversity consequences of economic sophistication are not primarily transmitted through changes in domestic land use, agricultural intensification, conservation effort, or sectoral composition. Instead, the evidence is more consistent with the pollution-haven logic of Copeland & Taylor (1994, 2004), recently revisited for trade-driven leakage by Holladay et al. (2018), Abman et al. (2026), and Carattini et al. (2026): as economies become more sophisticated, environmentally intensive production is increasingly relocated across borders. In the context of biodiversity, however, relocation does not sever the ecological connection between importing and exporting countries. Because species ranges frequently span national boundaries, habitat conversion abroad remains relevant for the conservation status of species occurring at home. The bridge analysis provides direct evidence for this mechanism by showing that imports sourced from biome-sharing regions are associated with larger subsequent declines in domestic biodiversity.

These findings also extend the multi-regional input–output literature on biodiversity footprints (Lenzen et al. 2012, Moran & Kanemoto 2017, Marques et al. 2019, Wilting et al. 2017, Chaudhary & Kastner 2016, Irwin et al. 2022) and complement microeconomic evidence on the biodiversity costs of trade at the producer level (Dalheimer et al. 2024). Previous work has established that wealthy economies externalize a substantial share of the biodiversity pressure embodied in their consumption. Our results identify economic sophistication as an independent structural determinant of that displacement, conditional on income, and link the geography of embodied biodiversity loss to observed changes in countries' Red List Indices through shared species ranges.

Two recent contributions warrant direct comparison. Wiebe & Wilcove (2025) document outsourced deforestation across 24 developed economies and show that consumption in wealthy economies drives forest loss in their trading partners; our results identify economic sophistication, beyond income, as the structural determinant of this displacement and connect it to within-country biodiversity outcomes through species-range overlap. Liang et al. (2021) use shocks to local US economic activity driven by national military buildups and find that they reduce species abundance, diversity, and stability, with air pollution explaining roughly a third of the effect; our cross-country evidence is complementary in scope, identifying the same direction of pressure but isolating the role of national production structure rather than local economic shocks, and showing that the within-country relationship survives accounting for income, population density, and broader development covariates.

More broadly, the results illustrate a general limitation of production-based environmental accounting in an increasingly fragmented world economy. Economic sophistication alters not only what countries produce but also where the environmental consequences of consumption are realized. For pollutants whose damages are spatially diffuse, territorial accounting may remain informative. For biodiversity, whose damages are tied to local habitat conversion and species distributions, territorial indicators can become systematically misleading. The distinction is likely to matter well beyond biodiversity whenever environmental externalities are embodied in global value chains.

The policy implications follow directly from this geography. Territorial and production-based indicators can understate the biodiversity consequences associated with more sophisticated economies when a larger share of their consumption footprint is realized through habitat conversion abroad. The theoretical framework shows that correcting this imbalance requires accounting for the cross-border component of the externality, rather than pricing only the biodiversity damage borne within the country where conversion occurs. In principle, this can be achieved either through a consumption-based charge that reflects the full social cost of habitat conversion or through equivalent international transfers. These instruments are equivalent in their first-best allocation but differ in implementation and incidence. A charge levied at the point of conversion can internalize the full social cost without requiring bilateral attribution, with the economic incidence determined by market conditions. Bilateral transfers from consuming to producing countries, by contrast, assign the corrective payment directly according to the source of demand but require information on trade relationships and international coordination.

This latter approach is closer to existing mechanisms such as debt-for-nature swaps and REDD+ payments,[20] which channel resources toward countries where conservation and avoided habitat conversion take place. More broadly, the result supports a shift toward wealth- and footprint-based biodiversity accounting, consistent with Dasgupta (2021), and with calls for reform of agricultural support and trade policies that affect biodiversity impacts along international supply chains (Vergez et al. 2025). Biodiversity policy should therefore complement territorial indicators with measures that account for the biodiversity impacts embodied in final consumption.

6 Conclusion

This paper asked whether the structure of an economy, beyond its level of income, shapes biodiversity loss. The evidence consistently indicates that it does. Conditional on income, greater economic sophistication is associated with higher extinction risk within countries. This relationship is not explained by domestic land use or production intensity. Instead, economically sophisticated countries increas-

[20] REDD+ (Reducing Emissions from Deforestation and forest Degradation, plus the sustainable management of forests and the conservation and enhancement of forest carbon stocks) is a framework under the United Nations Framework Convention on Climate Change (UNFCCC) under which countries receive payments for measured reductions in deforestation.

ingly externalize the biodiversity footprint of their consumption through international trade, becoming net importers of extinction risk. Sophisticated economies are not becoming gentler on nature: their apparent cleanliness reflects territorial accounting rather than genuine environmental improvement, as sophistication relocates the geography of biodiversity loss rather than reducing it.

The paper makes three contributions. First, it identifies economic sophistication as a structural determinant of biodiversity loss, beyond the level of income. Second, it develops a trade model that unifies the paper's empirical findings within a single mechanism linking comparative advantage, habitat conversion, and cross-border species ranges. Third, it provides the first evidence that trade-embodied biodiversity pressure feeds back to domestic conservation outcomes through international species-range overlap.

Economic sophistication, beyond income, determines how much biodiversity pressure a country displaces abroad—refining a literature that has already established that biodiversity conservation cannot rely on territorial accounting alone, given the substantial cross-border footprint embodied in wealthy economies' consumption. Conservation frameworks based solely on domestic indicators therefore systematically understate the biodiversity impacts of sophisticated economies. Because biodiversity loss is embodied in international trade but transmitted through species whose ranges extend across national borders, effective conservation requires accounting systems that assign responsibility according to final consumption as well as production. In this sense, the geography of biodiversity loss has outgrown the geography of conventional environmental accounting.

A natural next step is to extend bilateral extinction-risk footprint accounts into a panel dataset. Constructing such a panel would allow the dynamic relocation of biodiversity pressure to be observed directly, permitting the range-overlap mechanism identified here to be tested within countries over time rather than only across countries. Such data would permit the framework developed here to trace, within countries and over time, how economic sophistication is reshaping the global geography of biodiversity loss.

References

Abman, R., Jenkins, H. & Lundberg, C. (2026), 'The limits of cross-border environmental policies: Trade diversion as leakage', *Journal of Environmental Economics and Management* **137**, 103298. doi: https://doi.org/10.1016/j.jeem.2026.103298.

Anselin, L. (1988), *Spatial Econometrics: Methods and Models*, Kluwer Academic Publishers, Dordrecht. doi: https://doi.org/10.1007/978-94-015-7799-1.

Asafu-Adjaye, J. (2003), 'Biodiversity loss and economic growth: A cross-country analysis', *Contemporary Economic Policy* **21**(2), 173–185. doi: https://doi.org/10.1093/cep/byg003.

BirdLife International & Handbook of the Birds of the World (2019), 'Bird species distribution maps of the world. version 2019.1', http://datazone.birdlife.org/species/requestdis.

BirdLife International & Handbook of the Birds of the World (2025), 'Bird species distribution maps of the world', https://datazone.birdlife.org/species/requestdis. Downloaded 2026, non-commercial research data-use agreement.

Boleti, E., Garas, A., Kyriakou, A. & Lapatinas, A. (2021), 'Economic complexity and environmental performance: Evidence from a world sample', *Environmental Modeling & Assessment* **26**, 251–270. doi: https://doi.org/10.1007/s10666-021-09750-0.

Brooks, T. M., Mittermeier, R. A., Mittermeier, C. G., da Fonseca, G. A. B., Rylands, A. B., Konstant, W. R., Flick, P., Pilgrim, J., Oldfield, S., Magin, G. & Hilton-Taylor, C. (2002), 'Habitat loss and extinction in the hotspots of biodiversity', *Conservation Biology* **16**(4), 909–923. doi: https://doi.org/10.1046/j.1523-1739.2002.00530.x.

Butchart, S. H. M., Akçakaya, H. R., Chanson, J., Baillie, J. E. M., Collen, B., Quader, S., Turner, W. R., Amin, R., Stuart, S. N. & Hilton-Taylor, C. (2007), 'Improvements to the red list index', *PLoS ONE* **2**(1), e140. doi: https://doi.org/10.1371/journal.pone.0000140.

Butchart, S. H. M., Stattersfield, A. J., Bennun, L. A., Shutes, S. M., Akçakaya, H. R., Baillie, J. E. M., Stuart, S. N., Hilton-Taylor, C. & Mace, G. M. (2004), 'Measuring global trends in the status of biodiversity: Red list indices for birds', *PLoS Biology* **2**(12), e383. doi: https://doi.org/10.1371/journal.pbio.0020383.

Carattini, S., Huang, H., Pisch, F. & Singh, T. P. (2026), 'Trade and the scopes of pollution: Evidence from China's world market integration', *Journal of Environmental Economics and Management* **139**, 103378. doi: https://doi.org/10.1016/j.jeem.2026.103378.

Chaudhary, A. & Kastner, T. (2016), 'Land use biodiversity impacts embodied in international food trade', *Global Environmental Change* **38**, 195–204. doi: https://doi.org/10.1016/j.gloenvcha.2016.03.013.

Cincotta, R. P., Wisnewski, J. & Engelman, R. (2000), 'Human population in the biodiversity hotspots', *Nature* **404**(6781), 990–992. doi: https://doi.org/10.1038/35010105.

Cole, M. A. (2004), 'Trade, the pollution haven hypothesis and the environmental kuznets curve: examining the linkages', *Ecological Economics* **48**(1), 71–81. doi: https://doi.org/10.1016/j.ecolecon.2003.09.007.

Copeland, B. R. & Taylor, M. S. (1994), 'North-south trade and the environment', *The Quarterly Journal of Economics* **109**(3), 755–787. doi: https://doi.org/10.2307/2118421.

Copeland, B. R. & Taylor, M. S. (2004), 'Trade, growth, and the environment', *Journal of Economic Literature* **42**(1), 7–71. doi: https://doi.org/10.1257/.42.1.7.

Dalheimer, B., Parikoglou, I., Brambach, F., Yanita, M., Kreft, H. & Brümmer, B. (2024), 'On the palm oil–biodiversity trade-off: Environmental performance of smallholder producers', *Journal of Environmental Economics and Management* **125**, 102975. doi: https://doi.org/10.1016/j.jeem.2024.102975.

Dasgupta, P. (2021), *The Economics of Biodiversity: The Dasgupta Review*, HM Treasury, London.

Díaz, S., Settele, J., Brondízio, E. S., Ngo, H. T. et al. (2019), 'Pervasive human-driven decline of life on earth points to the need for transformative change', *Science* **366**, eaax3100. doi: https://doi.org/10.1126/science.aax3100.

Dietz, S. & Adger, W. N. (2003), 'Economic growth, biodiversity loss and conservation effort', *Journal of Environmental Management* **68**(1), 23–35. doi: https://doi.org/10.1016/s0301-4797(02)00231-1.

Growth Lab at Harvard University (2024), 'The atlas of economic complexity'. Data: https://atlas.hks.harvard.edu.

Hartmann, D., Guevara, M. R., Jara-Figueroa, C., Aristarán, M. & Hidalgo, C. A. (2017), 'Linking economic complexity, institutions, and income inequality', *World Development* **93**, 75–93. doi: https://doi.org/10.1016/j.worlddev.2016.12.020.

Hausmann, R., Hidalgo, C. A., Bustos, S., Coscia, M., Simoes, A. & Yildirim, M. A. (2014), *The Atlas of Economic Complexity: Mapping Paths to Prosperity*, The MIT Press, Cambridge, MA. doi: https://doi.org/10.7551/mitpress/9647.001.0001.

Hausmann, R., Hwang, J. & Rodrik, D. (2007), 'What you export matters', *Journal of Economic Growth* **12**, 1–25. doi: https://doi.org/10.1007/s10887-006-9009-4.

Hidalgo, C. A. (2021), 'Economic complexity theory and applications', *Nature Reviews Physics* **3**, 92–113. doi: https://doi.org/10.1038/s42254-020-00275-1.

Hidalgo, C. A. & Hausmann, R. (2009), 'The building blocks of economic complexity', *Proceedings of the National Academy of Sciences* **106**, 10570–10575. doi: https://doi.org/10.1073/pnas.0900943106.

Hidalgo, C. A., Klinger, B., Barabási, A.-L. & Hausmann, R. (2007), 'The product space conditions the development of nations', *Science* **317**, 482–487. doi: https://doi.org/10.1126/science.1144581.

Hoffmann, M., Hilton-Taylor, C., Angulo, A., Böhm, M., Brooks, T. M., Butchart, S. H. M., Carpenter, K. E., Chanson, J., Collen, B., Cox, N. A. et al. (2010), 'The impact of conservation on the status of the world's vertebrates', *Science* **330**(6010), 1503–1509. doi: https://doi.org/10.1126/science.1194442.

Holladay, J. S., Mohsin, M. & Pradhan, S. (2018), 'Emissions leakage, environmental policy and trade frictions', *Journal of Environmental Economics and Management* **88**, 95–113. doi: https://doi.org/10.1016/j.jeem.2017.10.004.

IPBES (2019), *Global Assessment Report on Biodiversity and Ecosystem Services*, IPBES Secretariat, Bonn, Germany. doi: https://doi.org/10.5281/zenodo.6417333.

Irwin, A., Geschke, A., Brooks, T. M., Siikamäki, J., Mair, L. & Strassburg, B. B. N. (2022), 'Quantifying and categorising national extinction-risk footprints', *Scientific Reports* **12**(1), 5861. doi: https://doi.org/10.1038/s41598-022-09827-0.

IUCN (2012), *IUCN Red List Categories and Criteria: Version 3.1*, second edn, IUCN, Gland, Switzerland and Cambridge, UK.

IUCN (2024), 'The IUCN Red List of threatened species: Spatial data (digital distribution maps)', https://www.iucnredlist.org/resources/spatial-data-download.

Lenzen, M., Moran, D., Kanemoto, K., Foran, B., Lobefaro, L. & Geschke, A. (2012), 'International trade drives biodiversity threats in developing nations', *Nature* **486**(7401), 109–112. doi: https://doi.org/10.1038/nature11145.

Li, X. & Chen, S. (2024), 'Does trade openness aggravate embodied species loss? evidence from the belt and road countries', *Environmental Impact Assessment Review* **104**, 107343. doi: https://doi.org/10.1016/j.eiar.2023.107343.

Liang, Y., Rudik, I. & Zou, E. (2021), Economic activity and biodiversity in the united states, NBER Working Paper 29357, National Bureau of Economic Research. doi: https://doi.org/10.3386/w29357.

Marques, A., Martins, I. S., Kastner, T., Plutzar, C., Theurl, M. C., Eisenmenger, N., Huijbregts, M. A. J., Wood, R., Stadler, K., Bruckner, M., Canelas, J., Hilbers, J. P., Tukker, A., Erb, K. & Pereira, H. M. (2019), 'Increasing impacts of land use on biodiversity and carbon sequestration driven by population and economic growth', *Nature Ecology & Evolution* **3**(4), 628–637. doi: https://doi.org/10.1038/s41559-019-0824-3.

Maxwell, S. L., Fuller, R. A., Brooks, T. M. & Watson, J. E. M. (2016), 'Biodiversity: The ravages of guns, nets and bulldozers', *Nature* **536**, 143–145. doi: https://doi.org/10.1038/536143a.

Mealy, P. & Teytelboym, A. (2022), 'Economic complexity and the green economy', *Research Policy* **51**, 103948. doi: https://doi.org/10.1016/j.respol.2020.103948.

Mills, J. H. & Waite, T. A. (2009), 'Economic prosperity, biodiversity conservation, and the environmental kuznets curve', *Ecological Economics* **68**, 2087–2095. doi: https://doi.org/10.1016/j.ecolecon.2009.01.017.

Moran, D. & Kanemoto, K. (2017), 'Identifying species threat hotspots from global supply chains', *Nature Ecology & Evolution* **1**, 0023. doi: https://doi.org/10.1038/s41559-016-0023.

Naidoo, R. & Adamowicz, W. L. (2001), 'Effects of economic prosperity on numbers of threatened species', *Conservation Biology* **15**(4), 1021–1029. doi: https://doi.org/10.1046/j.1523-1739.2001.0150041021.x.

Neagu, O. (2020), 'Economic complexity and ecological footprint: Evidence from the most complex economies in the world', *Sustainability* **12**, 9031. doi: https://doi.org/10.3390/su12219031.

Pandit, R. & Laband, D. N. (2007), 'Spatial autocorrelation in country-level models of species imperilment', *Ecological Economics* **60**(3), 526–532. doi: https://doi.org/10.1016/j.ecolecon.2006.07.018.

Pimm, S. L., Jenkins, C. N., Abell, R., Brooks, T. M., Gittleman, J. L., Joppa, L. N., Raven, P. H., Roberts, C. M. & Sexton, J. O. (2014), 'The biodiversity of species and their rates of extinction, distribution, and protection', *Science* **344**(6187), 1246752. doi: https://doi.org/10.1126/science.1246752.

Pimm, S. L. & Raven, P. (2000), 'Extinction by numbers', *Nature* **403**(6772), 843–845. doi: https://doi.org/10.1038/35002708.

Preston, F. W. (1962), 'The canonical distribution of commonness and rarity: Part I', *Ecology* **43**(2), 185–215. doi: https://doi.org/10.2307/1931976.

Romero, J. P. & Gramkow, C. (2021), 'Economic complexity and greenhouse gas emissions', *World Development* **139**, 105317. doi: https://doi.org/10.1016/j.worlddev.2020.105317.

Sobel, M. E. (1982), 'Asymptotic confidence intervals for indirect effects in structural equation models', *Sociological Methodology* **13**, 290–312. doi: https://doi.org/10.2307/270723.

Stern, D. I. (2004), 'The rise and fall of the environmental kuznets curve', *World Development* **32**, 1419–1439. doi: https://doi.org/10.1016/j.worlddev.2004.03.004.

Stojkoski, V., Koch, P. & Hidalgo, C. A. (2023), 'Multidimensional economic complexity and inclusive green growth', *Communications Earth & Environment* **4**, 130. doi: https://doi.org/10.1038/s43247-023-00770-0.

The World Bank (2020), *World Development Indicators*, The World Bank, Washington, DC.

United Nations Development Programme (2020), *Human Development Report 2020: The Next Frontier — Human Development and the Anthropocene*, United Nations Development Programme, New York. doi: https://doi.org/10.18356/9789210055161.

Vergez, A., Siikamäki, J., Mittempergher, D. & Piaggio, M. (2025), Agricultural support, biodiversity, and trade, Technical report, International Union for Conservation of Nature (IUCN). doi: https://doi.org/10.2305/rihu4909.

Vincent, J. R. (1997), 'Testing for environmental kuznets curves within a developing country', *Environment and Development Economics* **2**(4), 417–431. doi: https://doi.org/10.1017/s1355770x97000223.

Wiebe, R. A. & Wilcove, D. S. (2025), 'Global biodiversity loss from outsourced deforestation', *Nature* **639**, 389–394. doi: https://doi.org/10.1038/s41586-024-08569-5.

Wilting, H. C., Schipper, A. M., Bakkenes, M., Meijer, J. R. & Huijbregts, M. A. J. (2017), 'Quantifying biodiversity losses due to human consumption: A global-scale footprint analysis', *Environmental Science & Technology* **51**, 3298–3306. doi: https://doi.org/10.1021/acs.est.6b05296.

Online Appendix

Outsourcing Extinction: Economic Sophistication and the Geography of Biodiversity Loss

Komla Avoumatsodo Matías Piaggio Juha Siikamäki

A Derivations for the theoretical framework

This appendix collects the derivations underlying the propositions, lemmas, and key equations of Section 2.

A.1 Proof of Lemma 1 (composition effect)

Cost minimization in country i equates the value marginal product of labor across sectors,

$$\kappa_i \eta (L_i^y)^{\eta-1} = pB\eta (L_i^x)^{\eta-1}, \tag{A.1}$$

which gives the labor ratio

$$\frac{L_i^y}{L_i^x} = \left(\frac{\kappa_i}{pB}\right)^{1/(1-\eta)}. \tag{A.2}$$

Imposing the labor constraint $L_i^x + L_i^y = L_i$ delivers the labor share L_i^x/L_i as in Equation (2.10), and hence the level $L_i^x = L_i \cdot (L_i^x/L_i)$, since L_i is fixed. Differentiating at fixed p,

$$\frac{\partial L_i^x}{\partial \kappa_i} = -\frac{L_i}{\left[1+(\kappa_i/pB)^{1/(1-\eta)}\right]^2} \cdot \frac{(\kappa_i/pB)^{\eta/(1-\eta)}}{(1-\eta)\,pB} < 0. \tag{A.3}$$

Since $x_i = B(L_i^x)^\eta$ and $h_i = \gamma_i x_i$,

$$\frac{\partial x_i}{\partial \kappa_i} = B\eta (L_i^x)^{\eta-1}\frac{\partial L_i^x}{\partial \kappa_i} < 0, \qquad \frac{\partial h_i}{\partial \kappa_i} = \gamma_i \frac{\partial x_i}{\partial \kappa_i} < 0, \tag{A.4}$$

which is the statement of Lemma 1.

A.2 World price response and the signs of $dh_i/d\kappa_i$ and $dh_j/d\kappa_i$

Let $S(p,\kappa_i) \equiv x_i(p,\kappa_i) + x_{-i}(p)$ denote world supply of the habitat-intensive good, where $x_{-i}(p) \equiv \sum_{j\neq i} x_j(p,\kappa_j)$ aggregates the rest of the world's supply at fixed $\{\kappa_j\}_{j\neq i}$, and $D(p) \equiv \lambda \sum_k V_k/p$ world demand (from Cobb–Douglas, $c_k^x = \lambda V_k/p$ for every country k). World market clearing,

$$S(p,\kappa_i) = D(p), \tag{A.5}$$

defines the equilibrium price p implicitly. Differentiating at fixed incomes and $\{\kappa_j\}_{j\neq i}$ and applying the implicit function theorem,

$$\frac{dp}{d\kappa_i} = -\frac{\partial S/\partial \kappa_i}{\partial S/\partial p - \partial D/\partial p}. \tag{A.6}$$

The denominator is the standard excess-supply slope, positive under upward-sloping supply ($\partial S/\partial p = \partial x_i/\partial p + \sum_{j\neq i} \partial x_j/\partial p > 0$, each term positive by the same cost-minimization logic as Lemma 1 applied to p rather than κ) and downward-sloping demand ($\partial D/\partial p = -\lambda \sum_k V_k/p^2 < 0$). The numerator is negative because $\partial x_i/\partial \kappa_i < 0$ (Lemma 1). Hence

$$\frac{dp}{d\kappa_i} > 0, \tag{A.7}$$

the equilibrium relative price of the habitat-intensive good rises as country i grows more sophisticated. The implied movements in habitat conversion follow from $h_k = \gamma_k x_k$:

$$\frac{dh_i}{d\kappa_i} = \gamma_i \left[\frac{\partial x_i}{\partial \kappa_i} + \frac{\partial x_i}{\partial p}\frac{dp}{d\kappa_i}\right], \qquad \frac{dh_j}{d\kappa_i} = \gamma_j \frac{\partial x_j}{\partial p}\frac{dp}{d\kappa_i} > 0 \quad \text{for every } j \neq i. \tag{A.8}$$

The sign of $dh_j/d\kappa_i$ is immediate, since $\partial x_j/\partial p > 0$ and $dp/d\kappa_i > 0$. The sign of $dh_i/d\kappa_i$ is not: the bracket in Equation (A.8) nets the direct effect of i's own sophistication against the price feedback its own supply contraction sets off, and the two terms have opposite signs ($\partial x_i/\partial \kappa_i < 0$, but $\partial x_i/\partial p > 0$ and $dp/d\kappa_i > 0$). Showing $dh_i/d\kappa_i < 0$ therefore requires showing the direct effect dominates.

Because L_i^x/L_i in Equation (2.10) depends on κ_i and p only through the ratio $\kappa_i/(pB)$, differentiating it with respect to p by the same steps Appendix A.1 uses to differentiate it with respect to κ_i gives $\partial L_i^x/\partial p = -(\kappa_i/p)\,\partial L_i^x/\partial \kappa_i$, and hence, since $x_i = B(L_i^x)^\eta$,

$$\frac{\partial x_i}{\partial p} = -\frac{\kappa_i}{p}\frac{\partial x_i}{\partial \kappa_i}. \tag{A.9}$$

Substituting Equation (A.9) into the bracket in Equation (A.8) isolates the price feedback as an elasticity,

$$\frac{dh_i}{d\kappa_i} = \gamma_i \frac{\partial x_i}{\partial \kappa_i}[1-\varepsilon], \qquad \varepsilon \equiv \frac{\kappa_i}{p}\frac{dp}{d\kappa_i}, \tag{A.10}$$

so, since $\partial x_i/\partial \kappa_i < 0$, the direct effect dominates the price feedback, and $dh_i/d\kappa_i < 0$, exactly when $\varepsilon < 1$. Substituting Equation (A.9) into Equation (A.6) gives ε in terms of the same supply and demand slopes already used to sign $dp/d\kappa_i$,

$$\varepsilon = \frac{\partial x_i/\partial p}{\partial x_i/\partial p + \sum_{j\neq i} \partial x_j/\partial p + \lambda \sum_k V_k/p^2}. \tag{A.11}$$

Every term in the denominator of Equation (A.11) is non-negative, and $\lambda \sum_k V_k/p^2 > 0$ strictly, since aggregate income is always positive; the denominator therefore strictly exceeds the numerator, so $\varepsilon \in (0,1)$ regardless of how large a supplier i is in the world market for x. Even in the limiting case in which i is the world's only producer of the habitat-intensive good, so that $\sum_{j\neq i} \partial x_j/\partial p = 0$, downward-sloping world demand alone keeps $\varepsilon < 1$. Hence, from Equation (A.10),

$$\frac{dh_i}{d\kappa_i} < 0, \qquad \frac{dh_j}{d\kappa_i} > 0 \quad \text{for every } j \neq i. \tag{A.12}$$

Conversion migrates from i to every one of its trading partners as i grows more sophisticated, so that $dh_{-i}/d\kappa_i \equiv \sum_{j\neq i} dh_j/d\kappa_i > 0$ as well.

A.3 Derivation of the conditional effect (Eq. 2.12)

The conservation status of country i is $b_i = \sum_j \omega_{ij}(1-h_j/\bar{A}_j)^z$ (Equation (2.4)). At fixed income V_i and prices, h_i depends on κ_i directly, and h_j for every $j \neq i$ depends on κ_i through the common world price (Appendix A.2). Differentiating b_i with respect to κ_i,

$$\frac{db_i}{d\kappa_i} = -\sum_j \omega_{ij} \cdot \frac{z}{\bar{A}_j}\left(1-\tfrac{h_j}{\bar{A}_j}\right)^{z-1} \frac{dh_j}{d\kappa_i}. \tag{A.13}$$

Substituting the definition $m_j \equiv (z/\bar{A}_j)(1-h_j/\bar{A}_j)^{z-1} > 0$ from Equation (2.5) and separating the $j = i$ term from the sum over $j \neq i$ yields

$$\frac{db_i}{d\kappa_i} = -\,\omega_{ii}\, m_i \frac{dh_i}{d\kappa_i} - \sum_{j\neq i} \omega_{ij}\, m_j \frac{dh_j}{d\kappa_i}, \tag{A.14}$$

which is Equation (2.12). The first term is non-negative, since $dh_i/d\kappa_i \leq 0$ by Lemma 1 and Appendix A.2; every term in the sum is non-positive, since $dh_j/d\kappa_i \geq 0$ for $j \neq i$ by Appendix A.2. Equation (2.12) is negative exactly when the sum dominates the first term.

A.4 Derivation of Proposition I

Let $g_k \equiv (1-h_k/\bar{A}_k)^z$ denote the survival share in country k. From Equation (2.4),

$$b_i = \sum_k \omega_{ik}\, g_k. \tag{A.15}$$

Treating $\{g_k\}_{k=1}^N$ as random across countries with finite second moments, the covariance between any two countries' biodiversity levels is

$$\operatorname{Cov}(b_i, b_j) = \operatorname{Cov}\left(\sum_k \omega_{ik}\, g_k, \sum_l \omega_{jl}\, g_l\right) = \sum_{k,l} \omega_{ik}\, \omega_{jl} \operatorname{Cov}(g_k, g_l). \tag{A.16}$$

If, in addition, the $\{g_k\}$ are mutually non-negatively correlated (a property that holds when habitat conversion shocks are themselves non-negatively correlated across countries, which the data support), then every term in Equation (A.16) is non-negative, and the sum is strictly positive whenever there exists k such that $\omega_{ik}\, \omega_{jk} > 0$. This proves the covariance property in Proposition I.

For the spatial-autoregressive (SAR) representation, let $\mathbf{b} = (b_1, \ldots, b_N)^\top$ and construct the row-standardized weight matrix directly from Ω,

$$W_{ij} \;=\; \frac{\omega_{ij}}{\sum_{k\neq i}\omega_{ik}}, \quad j\neq i, \qquad W_{ii} \;=\; 0, \tag{A.17}$$

so each row of W sums to one. The best linear projection of b_i onto the spatial lag $(W\mathbf{b})_i = \sum_{j\neq i} W_{ij}\, b_j$ is, by the standard projection theorem,

$$b_i \;=\; \rho\,(W\mathbf{b})_i + \xi_i, \qquad \rho \;=\; \frac{\mathrm{Cov}(b_i,(W\mathbf{b})_i)}{\mathrm{Var}((W\mathbf{b})_i)}, \qquad E[\xi_i\,(W\mathbf{b})_i] = 0. \tag{A.18}$$

By Equation (A.16), the numerator of ρ is non-negative, since $W_{ij} > 0$ exactly when $\omega_{ij} > 0$, and the SAR coefficient $\rho \geq 0$ measures the share of b_i's variation attributable to its spatial neighbors. The residual ξ_i is orthogonal to the spatial lag by construction.

The choice of W is empirical: Table 6 estimates Equation (4.4) directly under the three species-range overlap measures Ω described in Appendix D.1, and, as a further check against the field's standard geographic default, under a queen-contiguity weight matrix, without affecting the qualitative conclusion that spatial dependence does not explain away the within-country sophistication–extinction-risk relationship.

A.5 Proof of Proposition IV (consumption-based instrument)

Planner's problem. The planner chooses $\{c^x_i, c^y_i, L^x_i, L^y_i\}_{i=1}^N$ to maximize $\mathscr{W}$ in Equation (2.16) subject to the technology $x_i = B(L^x_i)^\eta$ and $y_i = \kappa_i(L^y_i)^\eta$, the labor constraint $L^x_i + L^y_i = L_i$, the habitat-conversion identity $h_i = \gamma_i x_i$, the biodiversity measure $b_i = \sum_j \omega_{ij}(1-h_j/\bar{A}_j)^z$, and the goods-market clearing conditions. Substituting the definitional constraints into the objective yields the Lagrangian

$$\mathscr{L} \;=\; \sum_i n_i\big[\lambda\log c^x_i + (1-\lambda)\log c^y_i + \phi\, b_i\big] + \mu_x\left(\sum_i x_i - \sum_i c^x_i\right) + \mu_y\left(\sum_i y_i - \sum_i c^y_i\right), \tag{A.19}$$

where μ_x, μ_y are shadow world prices.

Planner's first-order conditions. Differentiating $\mathscr{L}$ with respect to L^x_j and using $\partial h_j/\partial L^x_j = \gamma_j B\,\eta\,(L^x_j)^{\eta-1}$,

$$\mu_x\cdot\frac{\partial x_j}{\partial L^x_j} \;=\; \mu_y\cdot\frac{\partial y_j}{\partial L^y_j} \;+\; \mathrm{SMC}_j\cdot\frac{\partial h_j}{\partial L^x_j}, \tag{A.20}$$

with SMC_j defined in Equation (2.17). The FOC with respect to c^x_k yields the Cobb–Douglas relation $\mu_x = n_k\,\lambda\,/\,c^x_k$, and similarly for c^y. The planner therefore equates the value marginal product of labor in x, net of the social marginal cost of conversion, to the value marginal product of labor in y.

Decentralized equilibrium. Firms in j choose L^x_j to maximize profit $p_x x_j - w_j L^x_j$, where w_j is the wage in country j, with FOC

$$p_x\cdot\frac{\partial x_j}{\partial L^x_j} \;=\; w_j, \tag{A.21}$$

ignoring SMC_j entirely. The wage equates across sectors, $w_j = p_y\,\partial y_j/\partial L^y_j$. A consumer in country k choosing c^x_k takes prices as given and perceives the private marginal cost of an extra unit as p_x, plus at most the part of conversion damage that falls on her own species, PMC_{kj} from Equation (2.18). The wedge $\Delta_{kj} = \mathrm{SMC}_j - \mathrm{PMC}_{kj} > 0$ whenever conversion in j affects species shared by countries other than k.

Pigouvian instrument. Let τ_j denote a per-unit charge on the embodied habitat that consumption causes to be converted in j. The consumer's modified FOC adds τ_j to the effective marginal price of consumption originating in j. Setting

$$\tau_j \;=\; \mathrm{SMC}_j \;=\; \phi \sum_i n_i\, \omega_{ij}\, m_j \tag{A.22}$$

adds the full social marginal damage to the consumer's price, so the consumer's modified FOC coincides with Equation (A.20). The decentralized allocation under τ therefore implements the planner's first-best.

Equivalent decentralization. A transfer of Δ_{kj} from k to j implements the first-best only when j has a single consuming country: k already internalizes PMC_{kj} privately, and $\mathrm{PMC}_{kj} + \Delta_{kj} = \tau_j$, so the transfer brings k's total effective cost up to the full social marginal cost. With multiple consumers, charging each one the full Δ_{kj} over-internalizes: every Δ_{kj} already contains nearly all of τ_j, so j would collect several times the actual externality.

Scaling each transfer by k's own share of j's output avoids this. Let x_j^k denote what country k consumes of j's production, $\theta_{kj} \equiv x_j^k / x_j$, so $\sum_k \theta_{kj} = 1$ over every consuming country, including j itself. Setting the transfer to $T_{kj} = \theta_{kj}\, h_j\, \Delta_{kj}$ and summing over k,

$$\sum_k T_{kj} \;=\; h_j \sum_k \theta_{kj}\big(\tau_j - \mathrm{PMC}_{kj}\big) \;=\; h_j \tau_j - h_j \underbrace{\sum_k \theta_{kj}\,\mathrm{PMC}_{kj}}_{\text{privately internalized}}\,, \tag{A.23}$$

using $\sum_k \theta_{kj} = 1$. The underbraced term is what is already privately internalized: each country's own private cost PMC_{kj}, weighted by its own share θ_{kj} of the conversion, summed across every consuming country. The transfers total $\tau_j h_j$ minus exactly this amount; adding the transfers back to it recovers the full $\tau_j h_j$, so the transfers replicate τ_j's incentive without over- or under-charging regardless of how many countries import from j. The $k = j$ term is a transfer from j to itself and nets to zero, so only cross-border transfers move in practice.

B Validity of the Complexity Outlook Index as an instrument

To assess the validity of the Complexity Outlook Index (COI) as an instrument for the ECI, we run a battery of tests, collected in Table B.1. First, instrument relevance is established by a strong first stage (coefficient 0.381, $p < 0.01$; Kleibergen–Paap rk Wald $F = 31.4$), well above conventional weak-instrument thresholds. Second, and most directly bearing on the exclusion restriction, when the COI is entered into the structural equation alongside the ECI its coefficient is small and statistically insignificant (-0.002, $p = 0.61$): the instrument has no effect on extinction risk other than through the ECI. Third, a falsification test using a lead of the COI returns a null effect (-0.003, $p = 0.33$), so the instrument does not pick up anticipation or pre-trends in the outcome.

Table B.1: Instrument-validity tests for the Complexity Outlook Index

Test	Statistic	Implication
(1) Relevance: first-stage coef. on COI	0.381^{***} (0.068)	strong first stage, $F = 31.4$
(2) Exclusion: COI $\mid$ ECI in structural eq.	-0.002 (0.004)	no direct effect
(3) Falsification: lead of COI	-0.003 (0.003)	no anticipation/pre-trend

Notes: Six-period panel, 1993–2019, with country and period fixed effects and the controls of Equation (3.4); standard errors clustered by country. Row (1) reports the first-stage coefficient of ECI on the COI, with its significance stars, and the corresponding F-statistic. Row (2) adds the COI to the structural equation alongside ECI; an insignificant coefficient supports the exclusion restriction. Row (3) replaces the COI with its one-period lead. Coefficients carry significance stars where significant; for rows (2) and (3), statistical insignificance is the outcome that supports instrument validity. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C Further robustness checks

This appendix reports additional robustness checks supporting the results in Section 4. It complements the sequential control additions and spatial dependence analysis reported in the main text by documenting sensitivity to sample composition, functional form, and temporal aggregation, by comparing alternative estimators of the sophistication–RLI relationship, and by reporting the full first-link tests for the domestic channel summarized in Section 4.2.

C.1 Sample, functional form, and temporal aggregation

Table C.1 reports four variants of the preferred specification in Equation (3.4). Column 1 reproduces the baseline six-period two-way fixed-effects specification with country and period fixed effects. Column 2 restricts the sample to the balanced panel of countries observed in all six periods. Column 3 augments the specification with a quadratic in log GDP per capita to test for a biodiversity Environmental Kuznets Curve. Column 4 re-estimates the model using a coarser aggregation into three periods (1993–95, 2005–07, and 2017–19).

Across all specifications, the coefficient on the Economic Complexity Index remains negative, statistically significant, and stable in magnitude. The balanced-panel restriction leaves the estimate essentially unchanged, indicating that differential attrition or sample composition does not drive the results. The EKC specification shows no evidence of nonlinear income effects: neither the linear nor the quadratic term in log GDP per capita is statistically significant, and the inclusion of the quadratic term does not alter the ECI coefficient. Temporal aggregation to three periods leaves the estimate similarly stable, indicating that the result is not sensitive to the frequency at which the Red List Index is observed.

Overall, Table C.1 shows that the baseline relationship is robust to alternative samples, functional forms, and temporal aggregation schemes.

Table C.1: Further robustness checks

	(1) Preferred	(2) Balanced panel	(3) EKC quadratic	(4) Three-period panel
Economic Complexity Index	-0.012***	-0.013***	-0.012***	-0.015***
	(0.004)	(0.004)	(0.004)	(0.005)
log GDP per capita	-0.020**	-0.033***	-0.059	-0.033***
	(0.008)	(0.009)	(0.044)	(0.009)
log GDP per capita squared			0.002	
			(0.002)	
log Population density	-0.030***	-0.044***	-0.025**	-0.041***
	(0.011)	(0.012)	(0.012)	(0.011)
Human Development Index	0.134**	0.179**	0.151**	0.189***
	(0.061)	(0.077)	(0.060)	(0.066)
Within R^2	0.418	0.446	0.420	0.445
Observations	1,000	774	1,000	484

Notes: Columns report the within-country specification Equation (3.4) with country and period fixed effects. Column 1 is the preferred six-period panel. Column 2 restricts the sample to countries observed in all six periods. Column 3 adds log GDP per capita squared to test the biodiversity Environmental Kuznets Curve; neither the linear nor the quadratic income term is statistically significant. Column 4 re-estimates the model on three longer periods (1993–95 / 2005–07 / 2017–19) instead of the preferred six-period panel. Standard errors clustered by country in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C.2 First-link tests for the domestic channel

In this section, we test a set of candidate channels through which ECI could affect RLI domestically, including the possibility that greater sophistication raises demand for land-intensive inputs. A necessary condition for any such domestic channel is that economic sophistication systematically affects the candidate mediator. We therefore test this directly by estimating, for each candidate channel X_{it},

$$X_{it} = \pi_0 + \pi_1 ECI_{it} + \pi_2 \log GDP_{it} + \pi_3 \log Popden_{it} + \pi_4 HDI_{it} + \theta_i + \delta_t + u_{it}, \tag{C.1}$$

on the six-period panel with country-clustered standard errors. The coefficient π_1 captures whether the ECI is associated with systematic changes in the proposed domestic channel, conditional on income, population density, and human development. A coefficient statistically indistinguishable from zero indicates that the candidate mediator does not respond systematically to ECI, ruling out the channel as a plausible first-stage mechanism. The absence of such domestic responses shifts attention toward the trade-embodied mechanism through which greater sophistication may displace biodiversity pressure across borders.

We consider variables that span the main theoretically plausible domestic transmission margins: land-use outcomes (forest cover, agricultural land, urban land, and the rate of deforestation), proximate drivers of habitat pressure (agricultural productivity, energy consumption, foreign direct investment, and protected-area coverage), and sectoral composition (value-added shares of manufacturing, services, and agriculture). Together, these variables cover, to the best of our knowledge, the principal domestic

Table C.2: First-link tests: effect of the ECI on candidate domestic channels

	(1) Forest	(2) Cropland	(3) Urban	(4) Deforest.	(5) AgTFP	(6) Energy	(7) FDI	(8) Prot. area	(9) Manuf.	(10) Serv.	(11) Agric.
Economic Complexity Index	0.0001	0.0058	0.0045	0.0950	0.0347	0.0713	1.8905	0.0025	0.0802*	0.0024	0.0160
	(0.0022)	(0.4936)	(0.0034)	(0.6519)	(0.0292)	(0.0513)	(1.2076)	(0.0025)	(0.0444)	(0.0193)	(0.0350)
Within R^2	0.002	0.124	0.021	0.023	0.035	0.231	0.008	0.080	0.018	0.009	0.099
Observations	997	999	997	993	920	997	998	808	934	950	974

Notes: Each column is a separate regression of the named candidate domestic channel on the standardized Economic Complexity Index, Equation (C.1). Columns: (1) Forest, forest share of land area; (2) Cropland, agricultural land share; (3) Urban, urban land share; (4) Deforest., the deforestation rate; (5) AgTFP, agricultural total factor productivity (log); (6) Energy, log energy consumption per capita; (7) FDI, foreign direct investment as a share of GDP; (8) Prot. area, protected-area share of land area; (9) Manuf., log manufacturing value-added share of GDP; (10) Serv., log services value-added share of GDP; (11) Agric., log agriculture value-added share of GDP. All regressions include country and period fixed effects, log GDP per capita, log population density, and the Human Development Index (coefficients omitted). Within R^2 and observation counts are reported for each channel so that precision, not just the point estimate, can be assessed directly; the smaller samples for columns 8 and 9 reflect coverage gaps in the underlying source data. Standard errors clustered by country in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

pathways through which changes in economic structure could plausibly affect biodiversity outcomes.

Ten of the eleven coefficients reported in Table C.2 are statistically indistinguishable from zero at conventional levels ($p > 0.10$). Conditional on income and the baseline controls, the ECI does not systematically affect forest cover, agricultural land, urban land, the rate of deforestation, agricultural productivity, energy consumption, foreign direct investment, protected-area coverage, or the sectoral shares of services and agriculture.

The only statistically detectable effect of the ECI is on the manufacturing value-added share, which increases modestly with sophistication (0.08, $p = 0.07$). This is consistent with the compositional mechanism embedded in the model (Lemma 1): as productive capabilities expand, labor is reallocated toward manufacturing and away from land-intensive activities. However, the relevant empirical question is whether this margin mediates the biodiversity relationship. Table 3 shows that it does not: including the manufacturing share in the baseline RLI specification leaves the coefficient on the ECI essentially unchanged at -0.013 (significant at the 1% level). Manufacturing responds to sophistication, but its magnitude is insufficient to account for the observed relationship with extinction risk.

A formal test of joint mediation. The first-link regressions above test only whether the ECI moves each candidate channel; eleven mostly null coefficients are weak evidence of absence on their own, since imprecision and true absence are observationally similar. They are also silent on how much of the ECI–RLI relationship any channel, even a moving one, could carry. Table C.3 closes both gaps with a classical product-of-coefficients mediation test (Sobel 1982) for each channel. Writing a for the first-stage coefficient on the ECI in Equation (C.1) and b for the channel's own coefficient when added to the baseline RLI regression, the indirect effect is $a \times b$, with standard error $\sqrt{b^2\,se(a)^2 + a^2\,se(b)^2}$ by the delta method. Both legs, and the corresponding total ECI effect, are estimated on the identical sample for each channel, so that the total effect equals the direct effect (the ECI coefficient with the channel included) plus the indirect effect exactly, and the mediated share is well defined.

Table C.3: Sobel mediation test: indirect effect of the ECI on RLI through each channel

Channel	First stage a	Second stage b	Indirect $a \times b$	Sobel z	Mediated share
Forest share	0.0001(0.0022)	0.2867(0.1397)	0.00004	0.07	-0.3%
Agricultural land share	0.0058(0.4936)	0.0004(0.0005)	0.00000	0.01	-0.0%
Urban land share	0.0045(0.0034)	$-0.0989(0.0607)$	-0.00045	-1.03	3.7%
Deforestation rate	0.0950(0.6519)	0.0001(0.0002)	0.00001	0.14	-0.1%
Agricultural TFP (log)	0.0347(0.0292)	$-0.0005(0.0070)$	-0.00002	-0.07	0.2%
log Energy consumption	0.0713(0.0513)	$-0.0117(0.0051)$	-0.00083	-1.19	7.9%
FDI (% of GDP)	1.8905(1.2076)	$-0.0000(0.0001)$	-0.00003	-0.22	0.3%
Protected-area share	0.0025(0.0025)	0.1338(0.0734)	0.00033	0.87	-3.5%
log Manufacturing VA share	0.0802(0.0444)	0.0062(0.0044)	0.00050	1.12	-4.1%
log Services VA share	0.0024(0.0193)	0.0081(0.0071)	0.00002	0.12	-0.2%
log Agriculture VA share	0.0160(0.0350)	$-0.0014(0.0050)$	-0.00002	-0.24	0.2%

Notes: a is the first-stage coefficient on the ECI (channel on ECI and controls); b is the channel's coefficient when added to the baseline RLI regression (RLI on ECI, the channel, and controls); both on the identical channel-specific sample. Indirect effect is $a \times b$; Sobel z tests it against zero via the delta-method standard error. Mediated share is the indirect effect as a fraction of the total ECI effect (direct plus indirect) on that same sample. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

No channel's indirect effect is statistically distinguishable from zero ($|z| < 1.2$ throughout, including manufacturing, the one channel with a marginally significant first stage). Summing the absolute indirect effects across all eleven channels, without allowing any of them to offset one another, an explicit worst-case construction rather than a joint significance test, yields at most 18.9% of the baseline ECI coefficient. At least four-fifths of the estimated relationship between economic sophistication and within-country extinction risk is therefore not attributable, even under this conservative bound, to any combination of the domestic channels considered.

Overall, economic sophistication does not systematically affect any candidate domestic channel in a way that could account for the biodiversity relationship. The estimated relationship between economic sophistication and biodiversity loss therefore does not appear to operate through standard domestic land-use, production-intensity, or sectoral-composition margins, pointing instead to a trade-embodied mechanism through which biodiversity pressure is displaced abroad.

C.3 Cross-country and within-country evidence on the biodiversity Kuznets curve

This section compares the within-country effect of economic sophistication on biodiversity with the cross-country pattern documented in the Environmental Kuznets Curve (EKC) literature. Because between-country and within-country estimates exploit different sources of variation, they need not yield the same relationship. The comparison therefore asks whether the within-country effect documented here is consistent with, or reverses, the pattern typically reported in cross-country studies of biodiversity and income, and examines the source of this divergence.

Beside the preferred TWFE specification, we consider two additional estimators. First, we estimate a between-country specification using country means,

$$\overline{\log RLI}_i = \alpha_0 + \alpha_1 \overline{ECI}_i + \alpha_2 \overline{\log GDP}_i + \alpha_3 \overline{\log Popden}_i + \alpha_4 \overline{HDI}_i + v_i, \tag{C.2}$$

where $\overline{X}_i = T_i^{-1} \sum_t X_{it}$; this isolates cross-country variation averaged over time, the source of identification used by cross-sectional studies in the biodiversity–income literature. Second, we estimate a pooled OLS specification that stacks all observations,

$$\log RLI_{it} = \pi_0 + \pi_1 ECI_{it} + \pi_2 \log GDP_{it} + \pi_3 \log Popden_{it} + \pi_4 HDI_{it} + e_{it}. \tag{C.3}$$

which combines within- and between-country variation without controlling for time-invariant heterogeneity. We include it because it provides an intermediate benchmark between the pure cross-sectional estimator above and the preferred within-country specification, Equation (3.4).

Each estimator is also computed in an Environmental Kuznets Curve variant by adding $(\log GDP_{it})^2$. Because higher values of the Red List Index indicate lower extinction risk, the EKC hypothesis implies a U-shaped relationship between RLI and income (equivalently an inverted-U in extinction risk).

Table C.4 reports results for all six specifications. The relationship between economic sophistication and biodiversity risk differs sharply across estimators. The between-country specification yields a positive coefficient on the ECI (0.048, $p < 0.01$), and pooled OLS yields a similar positive estimate (0.041, $p < 0.01$). In contrast, the within-country fixed-effects estimator yields a negative and statistically significant coefficient (-0.012, $p < 0.01$). The sign reversal indicates that cross-sectional variation is dominated by between-country differences in development and historical patterns of land conversion, while within-country changes in sophistication are associated with rising extinction risk.

Adding the EKC specification does not alter this pattern. In the between and pooled specifications, the quadratic income term is positive and at least weakly significant, consistent with a cross-sectional EKC pattern. In the within-country specification, neither the linear nor the quadratic income term is statistically significant, and the ECI coefficient remains unchanged. This indicates that the EKC pattern does not emerge from within-country dynamics.

Taken together, this evidence is consistent with the cross-sectional confounding cautioned by Stern (2004): cross-sectional EKC patterns reflect compositional differences across countries rather than a within-country causal relationship between income and biodiversity outcomes.

Table C.4: Between-country, pooled, and within-country estimates of the sophistication–RLI relationship, with EKC variants

	Between		Pooled OLS		Within FE	
	(1)	(2)	(3)	(4)	(5)	(6)
Economic Complexity Index	0.048***	0.041***	0.041***	0.036***	-0.012***	-0.012***
	(0.013)	(0.013)	(0.010)	(0.010)	(0.004)	(0.004)
log GDP per capita	0.041**	-0.175	0.038**	-0.162**	-0.020**	-0.059
	(0.020)	(0.112)	(0.015)	(0.082)	(0.008)	(0.044)
log GDP per capita squared		0.011*		0.011**		0.002
		(0.006)		(0.004)		(0.002)
log Population density	-0.030***	-0.030***	-0.029***	-0.029***	-0.030***	-0.025**
	(0.006)	(0.006)	(0.006)	(0.006)	(0.011)	(0.012)
Human Development Index	-0.447***	-0.335**	-0.394***	-0.307**	0.134**	0.151**
	(0.158)	(0.167)	(0.122)	(0.127)	(0.061)	(0.060)
R^2	0.179	0.197	0.175	0.192	0.418	0.420
Observations	179	179	1,000	1,000	1,000	1,000

Notes: The dependent variable is the log Red List Index. Columns 1–2 estimate Equation (C.2) by the between-country estimator (regression on country means); columns 3–4 estimate the pooled OLS specification, Equation (C.3), with standard errors clustered by country; columns 5–6 estimate the within-country fixed-effects specification, Equation (3.4), with standard errors clustered by country. "Linear" columns specify income linearly; "EKC" columns add $(\log GDP)^2$ to test for the biodiversity Environmental Kuznets Curve. Because higher RLI denotes lower extinction risk, the EKC predicts a negative coefficient on log GDP combined with a positive coefficient on its square, generating a U-shape on the RLI (the mirror image of the conventional inverted-U on threat). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C.4 Isolating economic sophistication from income

Another natural concern is that the ECI coefficient in the baseline specification simply reflects the correlation between the ECI and income. In our six-period panel, the within-country correlation between the ECI and log GDP per capita is 0.06 — essentially zero once time-invariant country heterogeneity is absorbed. The Economic Complexity Index therefore varies within countries over time on a margin distinct from income.

Two additional specifications sharpen this point. Table C.5 reports the baseline (column 1) alongside two robustness variants that isolate the ECI effect from the income channel. Column 2 uses a residualized ECI. In a first step, the ECI is projected on log GDP per capita, log population density, the Human Development Index, and country and period fixed effects; the residual, denoted $ECI_{it}^{\perp}$, captures the

Table C.5: Isolating economic sophistication from income: residualization and interaction

	(1) Baseline	(2) Residualized ECI	(3) ECI × log GDP
Economic Complexity Index	-0.012***		-0.012***
	(0.004)		(0.004)
Residualized ECI		-0.012***	
		(0.004)	
ECI × log GDP (mean-centred)			0.001
			(0.003)
log GDP per capita	-0.020**		-0.019**
	(0.008)		(0.008)
log Population density	-0.030***		-0.029***
	(0.011)		(0.011)
Human Development Index	0.134**		0.134**
	(0.061)		(0.061)
Within R^2	0.418	0.399	0.418
Observations	1,000	1,000	1,000

Notes: All specifications use the six-period within-country panel with country and period fixed effects and standard errors clustered by country. Column 1 reproduces the preferred baseline, Equation (3.4). Column 2 replaces the Economic Complexity Index with $ECI^{\perp}_{it}$, the residual from projecting ECI on log GDP per capita, log population density, HDI, and country and period fixed effects; the specification then regresses $\log RLI_{it}$ on $ECI^{\perp}_{it}$ retaining the fixed effects. By the Frisch–Waugh–Lovell theorem, the coefficient equals the ECI coefficient in column 1. Column 3 adds the interaction $ECI_{it} \times (\log GDP_{it} - \overline{\log GDP})$; the ECI coefficient is therefore the marginal effect of the ECI at the sample mean income level, and the interaction tests whether the ECI effect varies with income. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

variation in the ECI orthogonal to income and the other controls. In a second step, the log Red List Index is regressed on this residualized ECI, retaining the fixed effects. By the Frisch–Waugh–Lovell theorem the coefficient on $ECI^{\perp}$ equals the ECI coefficient in the joint specification of column 1; reporting the two side by side makes explicit that the estimated relationship reflects ECI variation that is orthogonal to income.

Column 3 augments the baseline specification with the interaction $ECI_{it} \times (\log GDP_{it} - \overline{\log GDP})$, where $\overline{\log GDP}$ is the sample mean. With log GDP mean-centred, the main ECI coefficient is the effect of the ECI at the average income level; the interaction coefficient tests whether the sophistication–RLI relationship varies with income.

The estimates confirm that the baseline coefficient is not an artefact of the ECI–income correlation. The residualized ECI in column 2 delivers a coefficient of -0.012 ($p < 0.01$), identical to the baseline as the theorem predicts. The interaction in column 3 is statistically indistinguishable from zero (0.001, $p = 0.78$), so the sophistication–RLI relationship does not vary with income within the sample.

C.5 Isolating economic sophistication from trade openness

The Economic Complexity Index and the extinction-risk footprint of Section 4.3 are constructed from different pipelines (ECI from UN Comtrade goods-trade data via the Growth Lab's method; the footprint from the 2013 Eora multi-region input–output table), but both ultimately trace to international trade-flow statistics, so a concern is that the leakage results in Table 4 partly reflect gross trade volume rather than the sophistication of a country's production. Table C.6 re-estimates Equation (4.1) controlling for log trade openness, measured as trade as a share of GDP in 2013 (The World Bank 2020). The specification uses the 154 countries with non-missing trade-openness data, reducing the sample from 173 countries.

Table C.6: Isolating economic sophistication from trade openness

	(1) log Imported	(2) Foreign share	(3) Net import
Economic Complexity Index	0.535***	0.117***	637.750***
	(0.143)	(0.025)	(221.672)
log GNI per capita	0.782***	0.104***	370.120*
	(0.139)	(0.025)	(220.453)
log Trade openness (% GDP)	-1.381***	0.143***	-690.227
	(0.253)	(0.039)	(605.682)
R^2	0.496	0.494	0.181
Observations	154	154	154

Notes: Re-estimates Equation (4.1) with log trade openness (World Bank, trade as a share of GDP, 2013) (The World Bank 2020) added as a control. Compare to Table 4: the ECI coefficient is essentially unaffected (0.535 vs. 0.487 on imported footprint; 0.117 vs. 0.123 on foreign share; 638 vs. 634 on net import). Heteroskedasticity-robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Trade openness itself enters significantly on imported footprint and foreign share, confirming it captures real variation, but conditioning on it does not displace the ECI coefficient on any outcome; if anything, the imported-footprint estimate strengthens and gains precision. Economic sophistication is not standing in for how much a country trades.

D Construction of the species-range overlap proxy and bridge variables

This appendix details the construction of the species-range overlap matrix $\Omega = (\omega_{ij})_{1\leq i,j\leq N}$ and the bridge variables that enter Equation (4.3) and Table 5.

D.1 The species-range overlap proxy ω_{ij}

Data sources. Ω is constructed directly from species range polygons rather than from a biogeographic classification. We use digital range maps for the four taxonomic groups that comprise the Red List Index itself: mammals, amphibians, and reef-building corals from the IUCN Red List spatial data portal (IUCN 2024), and birds from BirdLife International's Birds of the World distribution database (BirdLife International & Handbook of the Birds of the World 2025). Country boundaries are the Natural Earth 1:10m admin-0 (country-level) boundaries; the coarser 1:110m resolution used elsewhere in the paper's spatial weight matrices omits a material number of small states (island nations in particular), so the finer boundary set is used here to avoid systematically dropping them from Ω.

Filtering. Each range polygon carries IUCN/BirdLife presence and origin codes. We retain only extant, probably extant, or possibly extant ranges (presence codes 1–3) that are native or reintroduced (origin codes 1–2), excluding extinct, presence-uncertain, introduced, and vagrant occurrences, so that Ω reflects each species' current native distribution.

Construction. For each species s and country i, presence is recorded if s's filtered range polygon intersects i's boundary. Let S_i denote the set of (taxon, species) pairs present in country i, pooling all four taxonomic groups. The primary measure is the asymmetric overlap

$$\omega_{ij} = \frac{|S_i \cap S_j|}{|S_i|}, \tag{D.1}$$

the share of country i's assessed species whose range also extends into country j; this is the direct empirical counterpart of the model's ω_{ij} in Equation (2.4), which is likewise asymmetric. On the 252-country sample for which range data are available, mean species richness per country is 701 (median 448), and the off-diagonal mean of Ω is 0.21.

Robustness to the overlap measure. We report two robustness checks against this primary construction directly in Table 5: a symmetric Sørensen measure, $\omega_{ij}^{\mathrm{Sor}} = 2|S_i \cap S_j|/(|S_i| + |S_j|)$, and the coarser same-continent indicator, $\omega_{ij}^{c} = \mathbf{1}_{\{c(i)=c(j)\}}$ where $c(\cdot)$ assigns each country to one of the seven continents in the Natural Earth admin_0 dataset.

D.2 Construction of the bridge variables

Let $M[i,j]$ denote the entry of the 2013 bilateral extinction-risk footprint matrix (Irwin et al. 2022), giving the extinction risk borne by source i's species on account of consumption in destination j. For each destination i we construct the three bridge variables that enter Table 5:

$$F_i^{\mathrm{imp}} = \sum_{k \neq i} M[k,i] \qquad \text{(imported footprint: own consumption falling abroad)}, \tag{D.2}$$

$$F_i^{\omega} = \sum_{k \neq i} \omega_{ik} M[k,i] \qquad \text{(overlap-weighted imported footprint)}, \tag{D.3}$$

$$\sigma_i = F_i^{\omega} / F_i^{\mathrm{imp}} \qquad \text{(structural overlap share of imports)}. \tag{D.4}$$

On the 185-country sample for which both the bilateral footprint matrix and Ω are available, F_i^{imp} averages roughly 4,052 extinction-equivalents with a median of 857, and σ_i averages 0.05 with a median of 0.05. The dependent variable in Table 5 is the long-difference $\Delta \log RLI_i$ between the 1993–95 and 2018–19 windows of the six-period panel; merging with this panel leaves the estimation sample at 127 countries.

D.3 Timing of the bilateral footprint matrix

The bilateral footprint matrix is measured only in 2013, while the long difference in Table 5 runs from 1993–95 to 2018–19; the footprint is implicitly treated as a stable stand-in for the trade structure driving biodiversity pressure across the full 25-year window, even though it predates part of what it explains. As a check, we re-estimate the domestic-channel-controlled specification restricted to the post-2013 long difference, from the 2013–15 window (the panel period containing the footprint's measurement year) to 2018–19, so the footprint no longer predates any of the outcome it explains, under each of the three Ω measures compared in Table 5's columns 2–4: the primary species-range measure, the symmetric Sørensen alternative, and the same-continent proxy. Columns 1–2 are restricted to the same 127-country sample as Table 5's column 2; column 3 inherits that same restriction on top of the continent proxy's own smaller coverage, leaving 120 countries, matching Table 5's column 4. Differences across columns therefore reflect only the choice of Ω, not differing country coverage or a differing window.

Table D.1: Timing robustness: post-2013 long difference (dependent variable: $\Delta \log RLI$)

	(1) Species range	(2) Sørensen	(3) Continent
Overlap share of imports (σ)	-0.114**	-0.092*	-0.043**
	(0.056)	(0.051)	(0.018)
R^2	0.260	0.193	0.306
Observations	127	127	120

Notes: Each column reproduces the domestic-channel-controlled specification of Table 5's columns 2–4, restricted to the 2013–15 to 2018–19 long difference so the footprint no longer predates any of what it explains. Controls: Δ ECI, $\log F_i^{\text{imp}}$, $\log F_i^{\text{dom}}$, $\log F_i^{\text{exp}}$, Δ log GDP per capita, Δ log population density, Δ HDI. Heteroskedasticity-robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

On this shorter window the overlap-share coefficient, the object of this test, retains its sign and remains significant under all three Ω measures. Economic sophistication is included only as a control here; its own effect on RLI is established separately in Section 4.1 on the full panel.

Indirect evidence: persistence of trade and sophistication structure. A direct check on whether the footprint matrix's own structure is stable over the panel is not possible: no bilateral extinction-risk footprint matrix has been constructed for any year besides 2013.[21] As an indirect check, we instead test whether a country's ECI ranking, the trade and export structure the footprint matrix is itself built from, is persistent across the panel. Table D.2 reports Spearman rank correlations of the Economic Complexity Index between period pairs. Adjacent five-year periods correlate at $\rho = 0.94$ to 0.97, and the two ends of the full 25-year window still correlate at $\rho = 0.82$. A country's position in the global sophistication hierarchy is therefore highly persistent, which supports treating the 2013 snapshot as informative about each country's durable trade structure across the window, rather than a transient configuration specific to that year.

[21] We searched for a multi-year alternative, including the annual (2001–2015) bilateral deforestation-footprint data underlying Wiebe & Wilcove (2025); that data is not publicly available, only a single cumulative 2001–2015 total per species per destination country is released, with no year dimension.

Table D.2: Persistence of ECI rankings across the panel

Period pair	Spearman ρ	N
1993–95 to 1998–2000	0.958	183
1998–2000 to 2003–05	0.946	189
2003–05 to 2008–10	0.939	190
2008–10 to 2013–15	0.940	191
2013–15 to 2018–19	0.971	192
1993–95 to 2018–19 (full window)	0.816	183
1993–95 to 2013–15 (pre-footprint half)	0.817	183
2013–15 to 2018–19 (post-footprint half)	0.971	192

Notes: Spearman rank correlation of the Economic Complexity Index between the indicated period pairs, six-period panel, 1993–2019. The bottom panel reports the full 25-year window (1993–95 to 2018–19) and the two halves bracketing the footprint's own 2013 measurement year, which falls in the fifth period (2013–15).